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Key Points:

- The response of an aqua-planet Quasi-Biennial Oscillation (QBO) to global warming is broadly consistent with comprehensive climate model results
- The aqua-planet QBO period and amplitude decrease when carbon dioxide is quadrupled and sea surface temperatures are warmed by 4 K
- The decreased QBO amplitude aligns with increased tropical upwelling, and the shorter period is a result of heightened gravity wave drag

Supporting Information:

Supporting Information may be found in the online version of this article.

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Understanding the Response of the Quasi-Biennial Oscillation to Carbon Dioxide and Sea Surface Temperature Increases in Aqua-Planet Simulations

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Abstract The Quasi-Biennial Oscillation (QBO) of the tropical stratospheric zonal winds influences tracer transport, the stratospheric polar vortex, and extra-tropical surface variability. However, the response of the QBO to increasing greenhouse gas concentrations is still uncertain. In this study, an idealized aqua-planet configuration of the upcoming Community Atmosphere Model version 7 (CAM7) is leveraged to understand the drivers of QBO changes in a future climate. The aqua-planet model does not have a seasonal cycle and uses prescribed, zonally symmetric sea surface temperatures (SSTs), simplifying the attribution of QBO trends to dynamical and radiative drivers. In an idealized climate change scenario with quadrupled carbon dioxide concentrations and increased (by 4 K) SSTs, the QBO period decreases by about 10 months due to increased resolved and parameterized gravity wave drag, and the lower stratospheric QBO amplitude decreases in tandem with tropospheric expansion and strengthened tropical upwelling. Little change in the QBO amplitude and period is observed for a quadrupling of carbon dioxide alone. The QBO response in the aqua-planet simulation mirrors that of its fully coupled counterpart, suggesting that aqua-planets could be a useful framework for deciphering inter-model differences in QBO projections.

Plain Language Summary In the stratosphere, the tropical zonal winds reverse direction every 14 months. This phenomenon is called the Quasi-Biennial Oscillation (QBO), and it influences global weather and climate. Although simulations of the QBO continue to improve, it remains unclear how the QBO will evolve in a warming climate. In this study, the QBO response to climate change is examined in an aqua-planet climate model. Like other idealized atmosphere models, the aqua-planet framework adopts a smooth surface with a simplified surface temperature profile and eliminates the seasonal cycle. On the other hand, the aqua-planet model retains all of the physical processes of a comprehensive atmosphere model. While the idealized nature of the aqua-planet makes it easier to understand connections between global warming and the QBO, its use of comprehensive physics ensures that insights from aqua-planet simulations can inform full-complexity modeling. Here, warming sea surface temperatures are found to be the primary driver of aqua-planet QBO changes, which include more frequent wind reversals and weaker winds overall. Since the aqua-planet model can reproduce some realistic features of the QBO and its response to climate change, it may be a useful tool for understanding disagreements between climate model predictions of the QBO.

1. Introduction

Rivaled in its regularity only by the seasonal and diurnal cycles, the Quasi-Biennial Oscillation (QBO) of the equatorial stratospheric zonal winds is an outstanding example of an atmospheric wave-mean flow interaction. The QBO consists of alternating bands of westerly and easterly winds that descend through the stratosphere at a speed of about one km per month and with an average period of 28 months (Baldwin et al., 2001; Naujokat, 1986). The amplitude, or intensity, of the QBO zonal winds is greatest in the equatorial middle stratosphere near an altitude of 30 km, decreasing away from the equator and toward the tropopause. A wide spectrum of vertically propagating equatorial waves generated by convective heating in the troposphere provides the momentum necessary to drive the descent of the QBO zonal jets (Dunkerton, 1997). These waves transfer their momentum to the mean flow either when their phase speeds equal the background wind speed (critical level absorption) or as a by-product of radiative damping (Holton & Lindzen, 1972). Studies of the QBO based on reanalysis products (Kim & Chun, 2015b; Pahlavan, Wallace, et al., 2021), satellite data (Alexander & Ortland, 2010; Ern et al., 2014), and climate model simulations (Garcia & Richter, 2019; Giorgetta et al., 2006; Krismer & Giorgetta, 2014) generally agree that the descending QBO westerlies are forced by equal parts planetary-scale Kelvin and small-scale gravity (SSG) waves, and that the descending easterlies are driven mainly by gravity wave forcing, with a smaller contribution from mixed Rossby-gravity (MRG) waves. Additionally, the descent of the QBO zonal winds can be slowed by tropical upwelling associated with the wave-driven meridional Brewer-Dobson circulation (BDC) (Kim, 2025; Match & Fueglistaler, 2020), which consists of a single overturning cell in each stratospheric hemisphere.

As the primary mode of variability in the tropical stratosphere, the QBO has far-reaching effects on the climate and variability of the stratosphere and troposphere. The vertical wind shear associated with the QBO induces a meridional temperature gradient and a corresponding secondary circulation cell (Anstey et al., 2022; Baldwin et al., 2001). The QBO secondary circulation modulates the transport of stratospheric ozone in the tropics, causing anomalously high ozone concentrations in the easterly QBO jet (Logan et al., 2003). In the extra-tropical stratosphere, the strength of the Northern Hemisphere winter polar vortex is closely related to the QBO phase (Holton & Tan, 1980). During the easterly phase of the QBO at 50 hPa, planetary-scale Rossby waves exert a drag on the mean flow, leading to a weaker and warmer vortex (Baldwin & Dunkerton, 2001). This weakening of the polar vortex biases the North Atlantic Oscillation toward its negative phase, which increases the likelihood of surface cold air outbreaks over Europe and eastern North America. Other tropospheric signatures of the QBO include strengthened tropical convection associated with the Madden-Julian Oscillation (MJO) and a poleward shift in the Pacific subtropical storm track when QBO winds are easterly at 50 hPa (Zhang & Zhang, 2018; J. Wang et al., 2018). In addition, there is evidence to suggest that the influence of the QBO on extra-tropical surface temperatures will be greater under climate change, highlighting the importance of accurate simulations of the future QBO (Rao et al., 2020).

Climate models still do not agree on how the QBO will react to increasing greenhouse gas forcing. Although simplified mechanistic models (Plumb, 1977; Saravanan, 1990; Takahashi & Boville, 1992) of the QBO suggest that increased momentum deposition by equatorial waves will decrease the period of the QBO, most general circulation models (GCMs) do not explicitly resolve SSG waves (Richter et al., 2020). Instead, the life cycle of small-scale waves is parameterized, meaning that it is estimated based on linear wave theory and the resolved atmospheric conditions. These parameterization schemes differ from model to model, are difficult to validate using observations, and are thus a large source of uncertainty in the modeled response of the QBO to a warmer climate. For example, the sign of the predicted change in the QBO period was inconsistent among the four Coupled Model Inter-comparison Project 5 (CMIP5) models that had a reasonable QBO (Kawatani & Hamilton, 2013). For CMIP6, a QBO was generated in a greater number (10) of participating models, but only seven displayed a statistically significant decrease in the QBO period (Butchart et al., 2020). The QBO initiative (QBOi), which analyzed 11 atmosphere-only climate models that could simulate a QBO, confirmed that the gravity wave momentum flux is negatively correlated with the QBO period in the lower stratosphere (Richter et al., 2022). However, the change in gravity wave momentum flux due to an abrupt increase in CO_2 was sensitive to the type of gravity wave parameterization used. Models that prescribed a fixed gravity wave source spectrum were unable to capture the evolution of the gravity wave forcing in a changing tropospheric climate and, as a result, predicted either no change in or a lengthening of the QBO period (Richter et al., 2022). In models that derived gravity wave properties from the simulated convection, the gravity wave flux changed due to global warming but there was no consensus on the sign of the QBO period change. Further complicating projections of the QBO period is the well-documented fluctuation of the descent rate of the QBO jets (Baldwin et al., 2001) due to the seasonal cycle in SSG wave momentum flux and tropical upwelling (Kim, 2025). For the QBOi models, the inter-cycle variability of the QBO period was also found to be highly dependent on the details of the gravity wave parameterizations (Bushell et al., 2022), likely contributing to inter-model discrepancies in the response of the QBO period to global warming.

Conversely, GCMs have been remarkably consistent in predicting global warming-induced weakening of the QBO amplitude in the lower stratosphere near 70 hPa. As greenhouse gas concentrations grow, the BDC is projected to strengthen (Butchart et al., 2006). Theoretical models of the QBO demonstrate that vertical advection– proportional to the vertical velocity in the tropical stratosphere– acts against the wave momentum flux and weakens the mean flow by expanding the QBO "buffer zone" (Match & Fueglistaler, 2020; Saravanan, 1990), which is the lower stratospheric region where the descent of the QBO jets is halted despite sufficient wave forcing (Match & Fueglistaler, 2019). Since the strengthening of the BDC is generally robust to model choice (e.g., Abalos et al., 2021; Butchart et al., 2006; Calvo & Garcia, 2009), a weakening of the QBO should also be expected. Indeed, all models participating in CMIP5, CMIP6, and the QBOi exhibit a decreased QBO amplitude in the lower stratosphere (below 40 hPa), especially during the easterly phase of the QBO (Butchart et al., 2020;

Kawatani & Hamilton, 2013; Richter et al., 2022). However, recent studies (Match & Fueglistaler, 2021; Oberländer-Hayn et al., 2016) have cast doubt on the causal link between BDC strength and QBO amplitude, arguing instead that both dynamical trends can be explained by an upward shift in stratospheric dynamics due to the expansion of the troposphere in a warmer climate. In contrast to robust changes in modeled QBO amplitude, significant trends in observed QBO amplitude remain elusive. While Kawatani and Hamilton (2013) identified a significant weakening trend in the QBO amplitude reconstructed from tropical radiosonde observations from 1953 to 2012, their results were skewed by record low QBO amplitudes from 2000 to 2012 (Match & Fueglistaler, 2021). Taking into account radiosonde observations between 2012 and 2021, Anstey et al. (2022) estimated a weakening trend in the QBO amplitude at 70 hPa of $3.5 \pm 3.0\%$ decade^{-1}, which is only about half of the estimate of Kawatani and Hamilton (2013). Match and Fueglistaler (2021) reported a similar statistically insignificant 3% decade^{-1} weakening trend, suggesting that any trends in QBO amplitude cannot yet be distinguished from interannual variability.

When fully coupled climate model predictions diverge, a hierarchy of idealized climate models can be useful for understanding the underlying dynamical mechanisms (Held, 2005). The National Center for Atmospheric Research (NCAR) Community Earth System Model (CESM) framework includes such a hierarchy, ranging from dry dynamical cores with simple physics (Held & Suarez, 1994), to aqua-planet models that replace land with a global ocean and use simplified (Frierson et al., 2006; Thatcher & Jablonowski, 2016) or full-complexity (Neale & Hoskins, 2000) physics schemes. Each of these models is designed to target a key source of uncertainty in modeling the climate system– sensitivity of the simulated climate to the choice of dynamical core, coupling between latent heat release and large-scale dynamics, and imperfect physical parameterizations, respectively. The full-physics aqua-planet model is the simplest climate model that can capture the interaction between radiation and greenhouse gases. At the same time, sources of external variability like the solar and seasonal cycles are removed, and the atmospheric lower boundary condition, a global ocean with temporally constant SSTs, is greatly simplified. Thus, an aqua-planet configuration of CESM is a valuable framework for understanding the interaction between greenhouse gas forcing and the drivers, especially those that are parameterized, of the QBO.

Until now, aqua-planet studies of the tropical stratosphere have largely focused on its temporal-mean response to climate change. In fact, one advantage of the aqua-planet climate's lack of external variability is that a mean climate state can be reached in only a few years of model time, as opposed to hundreds of years for a fully coupled climate model (Medeiros et al., 2016; Neale & Hoskins, 2000). Aqua-planet climate change experiments have confirmed and provided explanations for trends in atmospheric circulation seen in more complex models, such as the strengthening of the BDC (Eichelberger & Hartmann, 2005; Chen et al., 2010; S. Wang et al., 2012), the poleward shift in the tropospheric eddy-driven jet (Butler et al., 2010; Lu et al., 2010; Schemm & Röthlisberger, 2024), and the widening of the Hadley cell (Medeiros et al., 2015; Shaw & Tan, 2018). Additionally, aqua-planets often exhibit an equilibrium climate sensitivity similar to that of their parent model (Medeiros et al., 2008), suggesting that they will respond similarly to greenhouse gas radiative forcing. Although a QBO-like pattern has been generated in idealized climate models (Garfinkel et al., 2022; Horinouchi & Yoden, 1998; Yao & Jablonowski, 2013, 2015), this study is the first to analyze the response of the QBO to greenhouse gas forcing in an idealized model. Using the aqua-planet configuration of NCAR's Community Atmosphere Model version 7 (CAM7), this work will (a) quantify the response of the aqua-planet QBO and stratosphere to quadrupled well-mixed CO_2 and a 4 K increase in sea surface temperatures (SSTs), (b) identify pathways between the climate change-related temperature perturbations and the drivers of the QBO, and (c) compare the aqua-planet stratosphere response to that of more complex models.

The model setup, experimental details, and analysis methods are described in Section 2. Changes in the general circulation and their impact on the QBO amplitude, period and forcing terms are discussed in Section 3, and a summary of the main results and future work is presented in Section 4.

2. Model, Experiments, and Methods

2.1. Model

CAM7, the latest development version of the atmosphere component of CESM (Danabasoglu et al., 2020), uses the spectral element (SE) dynamical core (Dennis et al., 2012; Lauritzen et al., 2018) to solve the moist, hydrostatic primitive equations on a cubed-sphere grid and a suite of parameterization schemes to represent radiation, convection, cloud microphysics, turbulence, gravity waves, and chemistry. For CAM7, version 2 of the Morrison-Gettelman microphysics scheme (MG2) was replaced by the Parameterization of Unified Microphysics Across Scales, which introduces prognostic rimed ice and improved ice nucleation physics (Gettelman et al., 2023). Additionally, CAM7 uses a completely restructured version of the Rapid Radiative Transfer Model for General circulation applications (RRTMG: Mlawer et al., 1997; Iacono et al., 2008) that is more efficient than previous iterations. Recently discovered deficiencies in the CAM gravity wave parameterization scheme related to the calculation of the convective heating depth were also rectified. Although CAM7 received a new 93-level vertical grid designed to improve the representation of middle atmosphere dynamics (Simpson et al., 2025), its elevated model top (85 km) is positioned near the mesopause and lies well outside the QBO domain. As a result, a separate 72-level vertical grid from the Department of Energy's Energy Exascale Earth System Model version 2 (E3SM2; Golaz et al., 2022) was substituted for the 93 level grid in the present simulations to exclude the mesosphere (>0.01 hPa). The 72 level grid, with a model top at 0.12 hPa (60 km) (Xie et al., 2018), is still of high enough resolution in the stratosphere to resolve momentum deposition by waves (Richter et al., 2014). The interface level hybrid coefficients for the 72-level grid are included in Table S1 in Supporting Information S1. The SE dynamical core was run at the default ne30 resolution, which roughly corresponds to a 1° by 1° (110 km) horizontal grid point spacing.

The CAM7 aqua-planet configuration aligns with the Aqua-Planet Experiment (APE) protocol devised by Neale and Hoskins (2000). In order to remove complexity introduced by atmosphere-land interactions, the model surface is covered by an ocean that has a constant, zonally symmetric sea surface temperature (SST) profile. The analytic SST profile chosen for this study, called QOBS, approximates the observed annual-average SSTs. The QOBS profile peaks at $T_{\max} = 27°C$ at the equator, tapers to 0°C at the poles, and is given by:

$$SST(\lambda,\phi) = \begin{cases} \frac{T_{\max}}{2}\left[2 - \sin^2\left(\frac{3\phi}{2}\right) - \sin^4\left(\frac{3\phi}{2}\right)\right] & \phi \le \left|\frac{\pi}{3}\right| \\ 0 & \phi > \left|\frac{\pi}{3}\right| \end{cases} \tag{1}$$

where λ and ϕ symbolize the longitude and latitude (in radians), respectively.

The seasonal cycle was removed by setting the model's orbital parameters to their equinox values and applying a solar constant of 1,365 W/m^2. The lack of seasonal variability in the model circulation greatly simplifies the identification of greenhouse gas-induced climate changes. In accordance with the APE protocol, concentrations of all greenhouse gases except for water vapor, which is a prognostic quantity in the model, were prescribed, and aerosols were removed. Constant cloud ice and droplet numbers (1×10^6 m^{-3} and 0.1×10^6 m^{-3}, respectively) were applied in lieu of aerosol ice nucleation, as in a previous aqua-planet study using CAM (Medeiros et al., 2016).

CAM7 uses the Beres et al. (2005) scheme for parameterizing the generation of SSG waves by convective heating in the troposphere. The amplitude and phase speed distributions of gravity waves generated by the Beres et al. (2005) scheme are based on linear wave theory and depend on the depth of the convective heating, the mean wind within the heating region, and the maximum convective heating rate. Deeper convection generates faster gravity waves, and a non-zero mean wind skews the phase speed distribution toward positive or negative speeds. The amplitude of parameterized gravity wave momentum flux is proportional to the square of the maximum heating rate (Beres et al., 2002, 2004). In this study, the convective heating depth was scaled by a factor of 0.25 using the namelist parameter *gw_qbo_hdepth_scaling*. This is the default scaling used in the high-model top version of CESM, the Whole Atmosphere Community Climate Model (WACCM), and generates slower gravity waves that are more likely to break and deposit their momentum in the QBO (Gettelman et al., 2019; Lee et al., 2024). Such a scaling is needed because the Beres et al. (2005) scheme generates insufficient gravity wave momentum flux at low phase speeds compared to observations (Alexander et al., 2021). A large source of uncertainty in the Beres et al. (2005) scheme is the portion of the grid cell in which convection is occurring (Beres et al., 2005). Since this quantity is not an output of the CAM deep convection scheme, the convective fraction (CF) is a tunable parameter in the model. The amplitude of the gravity wave momentum flux is dependent on the convective fraction because it is used to scale the maximum convective heating rate. Thus, the CF parameter is usually set to obtain a realistic QBO period. Another tunable parameter in the Beres et al. (2005) scheme is the gravity wave efficiency, which is accessed through the namelist variable *effgw_beres_dp*. This parameter scales the amount of momentum deposited by parameterized gravity waves before it is applied to the mean flow, and can also be used to adjust the QBO period. The convective fraction and gravity wave efficiency were varied in tandem to produce a control aqua-planet QBO with a period as close to that of the observed QBO as possible. All relevant CAM7 parameters are specified in Appendix A.

2.2. Experiments

Three different experiments were conducted in this study to ascertain the impacts of increased atmospheric CO_2 on the aqua-planet QBO. The first (**1xCO2**) is a control simulation, which uses the APE default CO_2 concentration (348 ppmv) and the QOBS SST profile. The second experiment (**4xCO2**) explores the impact of quadrupled carbon dioxide (1,392 ppmv) on the climate. In addition to quadrupled CO_2, the third experiment (**4xCO2+4K**) applies a uniform 4 K increase to the QOBS SSTs to approximate surface warming beyond the stratospheric radiative adjustment timescale of about 1 month (Stjern et al., 2023). While an unrealistic future scenario, a quadrupling of CO_2 is often prescribed along with a 4 K increase in SSTs in both fully coupled (Cess et al., 1989; Richter et al., 2022) and aqua-planet (Medeiros et al., 2015) simulations to evaluate climate sensitivity. Conducting two elevated CO_2 simulations, one with increased SSTs and the other without, will reveal the extent to which the QBO is influenced by radiative perturbations in the stratosphere (i.e., from CO_2) or by dynamical coupling between the stratosphere and troposphere.

By design, the aqua-planet configuration removes external sources of climate variability. Thus, a stable aqua-planet climate can be obtained after only a couple of years in the model. However, in this study each experiment was run for 22 years to capture multiple complete QBO cycles. The first 2 years of each simulation were allocated to model spin-up from the analytic moist baroclinic wave initial conditions (Ullrich et al., 2014) in CAM and were not analyzed.

2.3. Methods

2.3.1. Determining QBO Amplitude and Period

The response of the QBO to increased greenhouse gas forcing can be quantified in terms of changes in its amplitude and period. The transition times method (Bushell et al., 2022) was used to calculate the QBO amplitude and period for each simulation. In this method, the zonal wind time series at 20 hPa is averaged from 4°S to 4°N and smoothed using a five-month running mean. This narrow tropical averaging region was chosen for the transition times method to account for fine meridional structure in the resolved wave forcing of the QBO, as in Garcia and Richter (2019) and Pahlavan, Wallace, et al. (2021). The time difference between consecutive westerly to easterly transitions in the 20 hPa zonal wind defines the period of a single QBO cycle, and the mean of this time difference across all cycles yields an average period. Similarly, the mean of the time series amplitude across all cycles yields an average QBO amplitude. These metrics were evaluated at 20 hPa for consistency with other modeling studies (Kim & Chun, 2015a; Krismer & Giorgetta, 2014). Dunkerton and Delisi (1985) noted that the amplitude of the QBO can also be approximated by $\sqrt{2}\sigma$, where σ is the standard deviation of the tropical zonal wind time series. Using this method, a vertical profile of the QBO amplitude was constructed for each case.

To circumvent the difficulty of comparing oscillations with different periods, the structure and drivers of the QBO will be analyzed only during its descending westerly and descending easterly phases. These QBO phases are identified as pairs of months when the zonal wind at 20 hPa changes sign (Kim & Chun, 2015a; Krismer & Giorgetta, 2014). A pair of months is classified as a descending westerly phase when the 20 hPa vertical wind shear is positive. This occurs when the QBO westerly jet is located above 20 hPa and the QBO easterly jet is located below 20 hPa. Similarly, a pair of months is classified as a descending easterly phase when the vertical wind shear at 20 hPa is negative. The model data are then averaged over all of the descending westerly and descending easterly phases for each 20-year experiment to create descending westerly and descending easterly composites. The split between descending westerly and descending easterly phases is necessary because the wind direction below 20 hPa determines which resolved waves can propagate into the stratosphere and deposit their momentum in the descending jet. As a result, the nature of the QBO drivers differs greatly between the two phases.

2.3.2. TEM Framework

A suitable framework for analyzing the drivers of the QBO is the Transformed Eulerian Mean (TEM) equations (Andrews et al., 1987). This is because the stratospheric climate is largely a result of small-amplitude waves interacting with the zonal-mean flow, with most stratospheric variability occurring in the meridional plane. The TEM equations are obtained by transforming the zonal-mean meridional velocity and vertical pressure velocity $(\bar{v}, \bar{\omega})$ as follows:

$$\bar{v}^* = \bar{v} - \frac{\partial \psi}{\partial p}, \tag{2a}$$

$$\bar{\omega}^* = \bar{\omega} + \frac{1}{a \cos \phi} \frac{\partial}{\partial \phi}(\psi \cos \phi). \tag{2b}$$

Here, $(\bar{v}^*, \bar{\omega}^*)$ is the TEM residual circulation in pressure coordinates with units (m s^{-1}, Pa s^{-1}), $a = 6.37123 \times 10^6$ m is the radius of Earth, p is atmospheric pressure, and

$$\psi = \overline{v'\theta'} / \frac{\partial \bar{\theta}}{\partial p} \tag{3}$$

is the eddy stream function with units kg s^{-3}. All values for physical constants in this study were taken from the Dynamics and Variability Model Inter-comparison Project protocol (Gerber & Manzini, 2016) to facilitate reproducibility. In Equation 3, θ is the potential temperature, zonal-mean terms are indicated with an overbar, and the primed quantities are deviations from the zonal mean. The two-dimensional residual flow can be concisely represented using the residual stream function (Abalos et al., 2021),

$$\overline{\Psi}^* = -\frac{\cos \phi}{g} \int_p^0 \bar{v}^* dp, \tag{4}$$

where $g = 9.80665$ m s^{-2} is the average gravitational acceleration at sea level. The residual stream function is used in this study as a proxy for the BDC, as in Abalos et al. (2021). The strength of the BDC is judged by calculating the total mass flux across a constant pressure surface in the tropical upwelling branch of the BDC (Abalos et al., 2021), which is bracketed by the turnaround latitudes. The Northern and Southern Hemisphere turnaround latitudes coincide with the maximum and minimum values, respectively, of the residual stream function on the constant pressure surface. Using the residual stream function extrema, $\overline{\Psi}^*_{\mathrm{max/min}}$, the upwelling mass flux at pressure level p is:

$$M(p) = 2\pi a \left(\overline{\Psi}^*_{\mathrm{max}}(p) - \overline{\Psi}^*_{\mathrm{min}}(p)\right). \tag{5}$$

When only the vertical component of the residual flow is considered, it will be transformed from a vertical pressure velocity $\bar{\omega}^*$ to a conventional vertical velocity $\bar{w}^*$ in log-pressure coordinates via a simple scaling,

$$\bar{w}^* = -\frac{R_d \bar{T}}{pg} \bar{\omega}^*, \tag{6}$$

where $R_d = 287.058$ J kg^{-1} K^{-1} is the ideal gas constant, $\bar{T}$ is the zonal average temperature, $R_d \bar{T}/g$ is a temperature-dependent atmospheric scale height, and the units of $\bar{w}^*$ are m s^{-1}. This scaling avoids inaccuracies in the vertical residual velocity that can be introduced by applying a constant scale height (Eichinger & Šácha, 2020).

The TEM zonal momentum budget equation listed below describes the dynamical sources of mean flow acceleration ($\bar{u}_t$) associated with the QBO:

$$\bar{u}_t = \left.\frac{\partial \bar{u}}{\partial t}\right|_{ADV} + \left.\frac{\partial \bar{u}}{\partial t}\right|_{EPFD} + \left.\frac{\partial \bar{u}}{\partial t}\right|_{GWD} + \left.\frac{\partial \bar{u}}{\partial t}\right|_{DIFF}. \tag{7}$$

The first term on the right hand side of Equation 7 represents the impact of advection (ADV) by the residual circulation and the Coriolis force on the QBO, and is defined as:

$$\left.\frac{\partial \bar{u}}{\partial t}\right|_{ADV} = \bar{v}^* \left[f - \frac{1}{a\cos\phi}\frac{\partial}{\partial \phi}(\bar{u}\cos\phi)\right] - \bar{\omega}^* \frac{\partial \bar{u}}{\partial p}, \tag{8}$$

where $f = 2\Omega \sin\phi$ is the Coriolis parameter and $\Omega = 7.29212 \times 10^{-5}\ \mathrm{s}^{-1}$ is the rotation rate of Earth. The second term on the right hand side of Equation 7 describes the zonal wind forcing due to planetary-scale wave drag, and is proportional to the divergence of the Eliassen-Palm (EP) flux:

$$\left.\frac{\partial \bar{u}}{\partial t}\right|_{EPFD} = \frac{\nabla \cdot \mathbf{F}}{a\cos\phi}. \tag{9}$$

The meridional and vertical components of the EP Flux, $\mathbf{F} = \left(0, F^{(\phi)}, F^{(p)}\right)$, are defined as:

$$F^{(\phi)} = a\cos\phi\left(\frac{\partial \bar{u}}{\partial p}\psi - \overline{u'v'}\right), \tag{10a}$$

$$F^{(p)} = a\cos\phi\left\{\left[f - \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}(\bar{u}\cos\phi)\right]\psi - \overline{u'\omega'}\right\}. \tag{10b}$$

Using these definitions, the zonal wind tendency due to planetary-scale wave drag can be constructed as follows:

$$\left.\frac{\partial \bar{u}}{\partial t}\right|_{EPFD} = \frac{1}{a\cos\phi}\left[\frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left(F^{(\phi)}\cos\phi\right) + \frac{\partial F^{(p)}}{\partial p}\right]. \tag{11}$$

An advantage of the TEM framework over the traditional Eulerian mean is that the influence of the wave-driven heat and momentum fluxes on the mean flow is contained in a single term (Equation 11). Climate simulations, however, cannot accurately model waves near or below the grid scale. Thus, the zonal wind tendency (body force) due to the Eliassen-Palm flux divergence in practice captures only the drag due to planetary-scale waves that are resolved by the model. Small-scale gravity waves are parameterized, and their impact on the zonal wind is included in a separate term, $\left.\frac{\partial \bar{u}}{\partial t}\right|_{GWD}$, which is diagnosed by the model. To close the zonal momentum budget, a residual term is calculated by subtracting the first three terms on the right hand side of Equation 7 from the total zonal wind tendency. This residual term contains all other unresolved sources of zonal acceleration, like numerical diffusion and radiative damping. However, in the aqua-planet simulations the residual zonal wind tendency is almost entirely due to explicit horizontal hyper-diffusion in the SE dynamical core (Lauritzen et al., 2018), and is labeled accordingly: $\left.\frac{\partial \bar{u}}{\partial t}\right|_{DIFF}$. An increase in the magnitude of the zonal wind acceleration, $\bar{u}_t$, in the descending westerly or descending easterly phase of the QBO is associated with a shorter QBO period, while a decrease in the magnitude of the zonal wind acceleration will lead to a longer QBO period.

2.3.3. Wave-by-Wave Breakdown

In nature, the QBO is forced by several different types of waves that are dynamically trapped in the equatorial region. These waves can be generated, for example, by latent heating from tropospheric deep convection and propagate vertically into the stratosphere. The meridional and vertical structure of equatorial waves can be deduced using shallow water theory (Lindzen, 1967; Matsuno, 1966) if the tropical atmosphere is approximated as a shallow water system with an equivalent depth, h. This theory admits four types of wave solutions:

eastward and westward propagating inertia-gravity (IG) waves, westward equatorial Rossby (ER) waves, eastward Kelvin waves, and westward MRG waves. In addition to the equivalent depth, h, three other parameters are needed to fully characterize these shallow water equatorial waves: the meridional mode number n, zonal wavenumber k, and frequency ω. The meridional mode number, zonal wavenumber, and frequency determine the meridional symmetry, spatial scale, and temporal scale, respectively, of an equatorial wave. Each equatorial wave type has a unique theoretical dispersion relation that relates the zonal wavenumber and frequency and can be used to calculate the wave's zonal phase speed. For shallow water Kelvin waves, the phase speed is $c_{p,x} = \sqrt{gh}$, whereas for other equatorial wave types the phase speed is a function of k and n. Eastward and westward-traveling equatorial waves respectively exert positive and negative wave forcings (Equation 11) on the QBO.

These vertically propagating waves can be identified in model data and observations using their characteristic spatiotemporal scales and (anti)symmetry with respect to the equator predicted by shallow water theory. In the method of Wheeler and Kiladis (1999), this is done by separating a model field U, like precipitation, into symmetric $U(\phi,\lambda)_{sym} \equiv \frac{1}{2}[U(\phi,\lambda) + U(-\phi,\lambda)]$ and antisymmetric $U(\phi,\lambda)_{asym} \equiv \frac{1}{2}[U(\phi,\lambda) - U(-\phi,\lambda)]$ parts, transforming the data into two-dimensional Fourier (wavenumber-frequency) space, and determining if the theoretical equatorial wave dispersion relationships overlap regions of significant spectral density, or power. Here, power is defined as the square of the model field in Fourier space, and significant power is obtained by dividing the symmetric and antisymmetric spectra by a smoothed background spectrum. Since only the time-average equatorial wave activity is of interest, these calculations are averaged over multiple overlapping (by 60 days) temporal segments with a length of 90 days. Significant tropical variability also exists outside of the spatiotemporal scales of the shallow water waves in observational (Wheeler & Kiladis, 1999) and model (Holt et al., 2022) data, and care should be taken to avoid misidentifying sources of variability where they overlap. For example, the MJO is an eastward-propagating area of organized convection that appears at low zonal wavenumbers and a frequency of 1/30 day^{-1} in the precipitation power spectrum (Wheeler & Kiladis, 1999). Although aqua-planet models generally cannot simulate the MJO, another type of propagating convective structure– the advective disturbance (AD)– is excited in the aqua-planet tropics (Nakajima et al., 2013; Rios-Berrios et al., 2020). ADs are westward-propagating antisymmetric disturbances with phase speeds between 2.5 m s^{-1} and 12 m s^{-1} and zonal wavenumbers between -28 and -4 (Nakajima et al., 2013). Spectral power from ADs partially overlaps the MRG dispersion curves in Fourier space.

Although the Wheeler and Kiladis (1999) method is useful for verifying the presence of equatorial waves and determining trends in equatorial wave power due to increasing carbon dioxide, it does not establish a connection between equatorial waves and the QBO. To accomplish this, the contribution of different wave types to the EP Flux divergence body force (Equation 11) must be determined. The approach in this study is to apply wavenumber-frequency ($k - \omega$) band-pass filters to the perturbation fields used in the calculation of the EP Flux divergence body force, as in Kim and Chun (2015b) and Pahlavan, Fu, et al. (2021). In this framework, negative zonal wavenumbers represent westward-propagating waves, and positive zonal wavenumbers represent eastward-propagating waves (Kim & Chun, 2015b). Kelvin waves were identified with $0 \le k \le 20$ and $\omega < 0.8$ day^{-1} in the symmetric spectrum. MRG waves were extracted from the antisymmetric spectrum for $|k| \le 20$ and $0.1 \le \omega \le 0.5$ day^{-1}. Thus, the MRG filter includes both westward- and eastward-propagating $n = 0$ inertia-gravity waves. The remaining wavenumber-frequency regions were classified as ER waves for $|k| \le 20$ and $\omega < 0.4$ day^{-1}, and IG waves for $|k| \le 20$ and $\omega \ge 0.4$ day^{-1}. Any perturbations with $|k| > 20$ were classified as SSG waves that are still large enough to be resolved by the model. These wavenumber-frequency ranges were chosen to represent the usual spatial and temporal scales of the equatorial waves derived from regions of significant power in the Wheeler and Kiladis (1999) method. Figure 1 shows the wavenumber-frequency filters in spectral space for the symmetric and antisymmetric components of perturbation zonal wind ($u' = u - \bar{u}$) and perturbation meridional wind ($v' = v - \bar{v}$). The shaded contours in Figure 1 represent regions of significant power in the lower stratosphere, and the black curves are theoretical dispersion relations for the four shallow-water equatorial wave types. Note that significant power occurs in the opposite component for meridional velocity than for zonal wind, potential temperature (not shown), and vertical pressure velocity (not shown), in agreement with the theoretical structure of the equatorial waves (Knippertz et al., 2022; Matsuno, 1966). There is no significant power for $-12 \le c \le -2.5$ m s^{-1} in the antisymmetric spectrum outside of the MRG dispersion curves (Figure 1b), suggesting that ADs are absent from the stratospheric perturbation fields. By enabling the

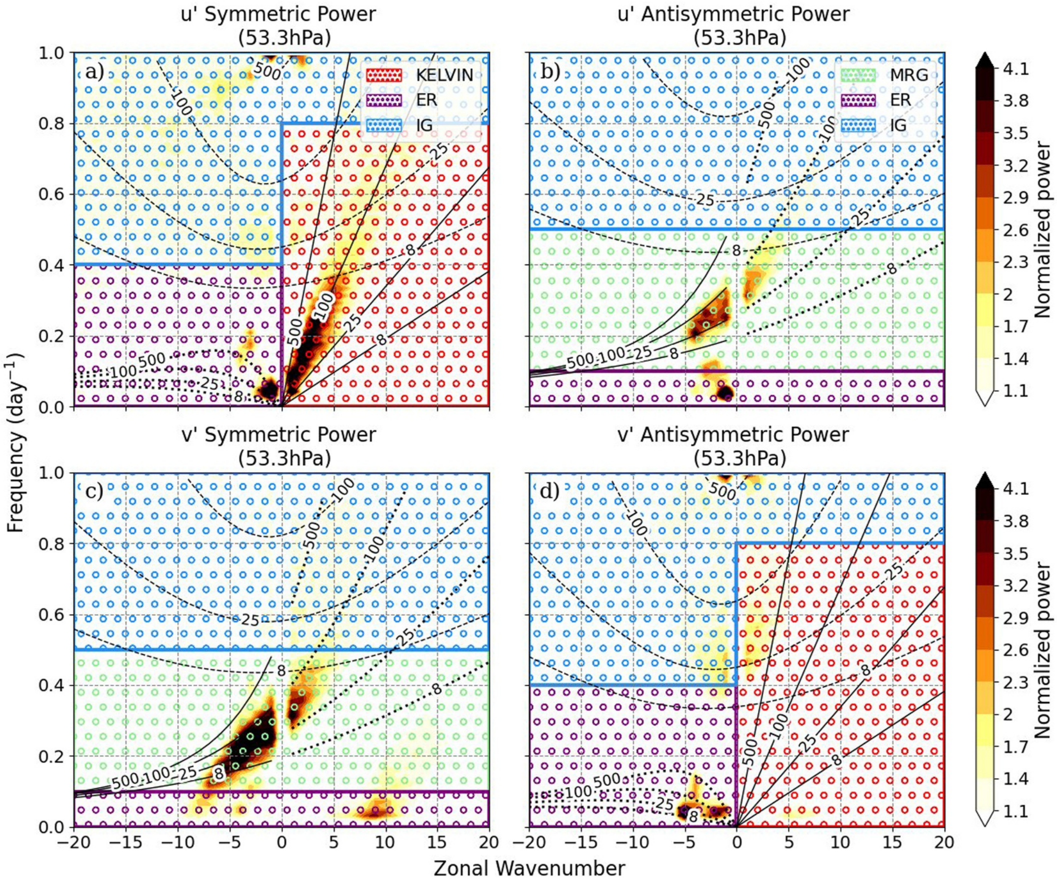

Figure 1. Significant (≥1.1) normalized power spectral density (shaded contours) for the symmetric (left) and antisymmetric (right) components of perturbation zonal wind u' (top) and perturbation meridional wind v' (bottom). The wavenumber frequency filters discussed in Section 2.3.3 are indicated by circular hatching. In the symmetric power spectrum of u', the solid black lines represent Kelvin wave dispersion relationships, the dotted lines equatorial Rossby waves, and the dashed lines $n = 1$ inertia-gravity (IG) waves. In the antisymmetric spectrum of u', the solid black lines represent mixed Rossby-gravity wave dispersion relationships, the dotted lines $n = 0$ IG waves, and the dashed lines $n = 2$ IG waves. The dispersion curves are labeled with equivalent depths in meters and are assigned to the opposite components for the perturbation meridional wind (v') power spectra.

robust identification of stratospheric equatorial waves in the model data, this analysis provides useful insight into how equatorial waves react to increasing carbon dioxide concentrations and SSTs and how these changes affect the QBO.

3. Results

Although the QBO is largely driven by resolved and parameterized wave drag, the mean circulation exerts an important influence on QBO changes in a climate with increased CO_2 and SSTs. Namely, background zonal wind conditions can modify the spectrum of waves that reach the stratosphere, and the BDC contributes to the advection term (ADV) in the QBO momentum budget (Equation 7). Thus, Sections 3.1 and 3.2 present and explain the mean circulation changes in the 4xCO2 and 4xCO2+4K experiments to provide context for a detailed discussion of the QBO momentum budget in Section 3.3.

3.1. Tropospheric Mean Circulation Response

Both of the elevated CO_2 experiments exhibit tropospheric warming due to increased well-mixed carbon dioxide. This response is shown in Figure 2, which displays control climatologies using black contours and climatological differences between the elevated CO_2 and control simulations in color for the zonal-mean zonal wind (a, b) and

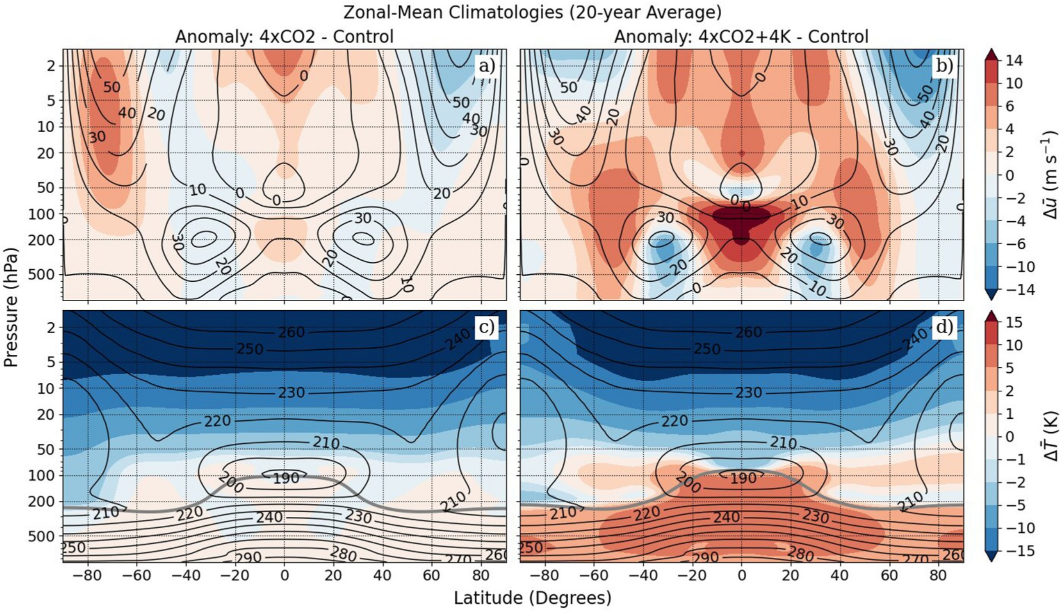

Figure 2. Zonal-mean differences between the 20-year control and elevated CO_2 climatologies. Shaded contours in (a) and (b) display zonal wind differences between the control and elevated CO_2 simulations, while the black contours are the control zonal winds. (c) and (d) are the same, but for temperature differences. The gray lines in (c) and (d) are the positions of the lapse-rate tropopause in the 4xCO2 and 4xCO2+4K simulations, respectively, according to the World Meteorological Organization definition.

temperature (c, d) fields. In the 4xCO2 experiment, the troposphere warms by less than 1 K everywhere (Figure 2c). In the 4xCO2+4K simulation, the tropical troposphere and polar troposphere warm over five times more than in the 4xCO2 simulation (Figure 2d), suggesting that the majority of the tropospheric warming in the aqua-planet stems from warmer SSTs. The warmed troposphere in the 4xCO2+4K simulation also expands vertically, with a tropical tropopause (solid gray line in Figure 2d) roughly 15 hPa higher than the control experiment. The tropopause pressure is calculated in the model using the World Meteorological Organization (WMO) algorithm described in Reichler et al. (2003).

Temperature anomalies in the aqua-planet atmosphere are accompanied by changes in the tropospheric mean circulation, such as a poleward shift in the mid-latitude zonal jet and an upward expansion of the subtropical zonal jet. The dipole pattern in the zonal-mean zonal wind anomalies centered at around 45°N/S in Figures 2a and 2b is indicative of a poleward shift of the maximum mid-latitude zonal winds. This shift in zonal wind is accompanied by a stronger absolute meridional temperature gradient on the poleward flank of the zonal jet in both hemispheres, obeying the thermal wind relationship. The upward shift in the subtropical jet appears in Figures 2a and 2b as positive zonal wind anomalies near the tropopause. The poleward shift of the mid-latitude zonal jet and the upward shift of the subtropical zonal jet are much more pronounced in the 4xCO2+4K simulation, owing to the increased thermal forcing associated with the warmer SSTs. A similar zonal wind response is observed in most idealized and comprehensive climate models (Butler et al., 2010; Chen et al., 2010; Richter et al., 2022). However, an unusually strong tropical upper tropospheric zonal jet appears near 150 hPa in the control zonal wind climatology (black contours in Figures 2a and 2b). A westerly zonal jet in the upper tropospheric tropics is a unique feature of aqua-planet simulations (Chien & Kim, 2024; Williamson et al., 2012), and appears to be driven by westerly equatorial Kelvin wave drag. The strengthening of the equatorial zonal winds near 150 hPa for the 4xCO2+4K case (Figure 2b) is thus a result of increasing Kelvin wave activity, which will be discussed in Section 3.3.1.

Another dynamical response of the aqua-planet atmosphere to elevated carbon dioxide and SSTs is the meridional expansion of the Hadley circulation cell. The Hadley cell expansion is visible in Figure 3, which shows the zonal-mean meridional wind control simulation climatology in black contours and the 4xCO2 and 4xCO2+4K anomalies in color (a,b), in addition to the zonal-mean total precipitation rate climatologies for all simulations (c).

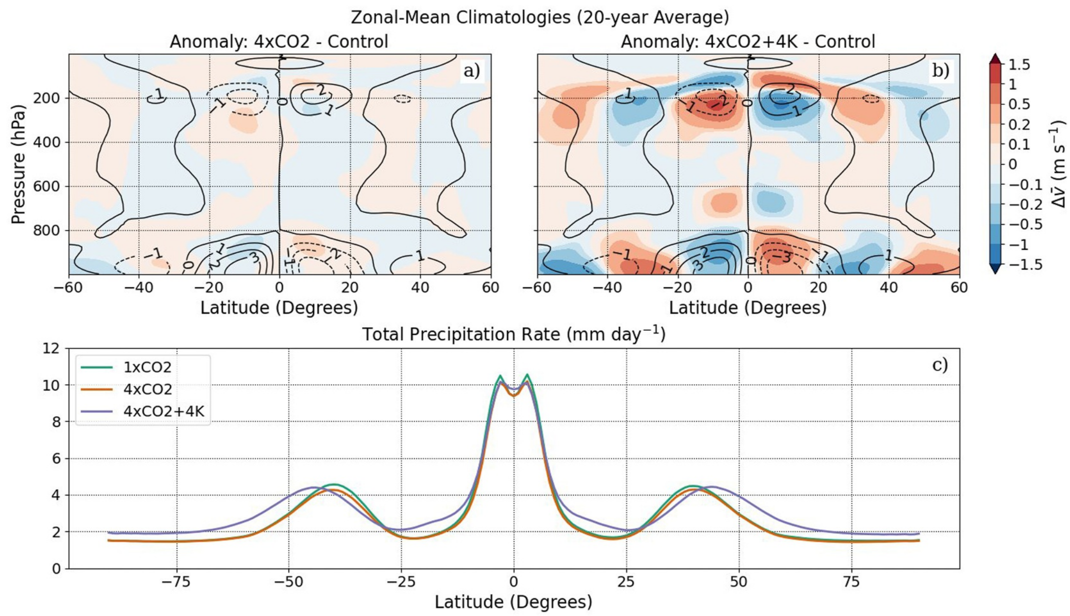

Figure 3. Zonal-mean, 20-year climatologies of meridional wind and total precipitation. The shaded contours in (a) and (b) represent differences in meridional wind between the control and elevated CO_2 simulations, and the black contours are the control meridional winds. (c) Shows the total precipitation rate for each simulation, as described in the legend.

For instance, the meridional wind near 25°N decreases in the 4xCO2+4K case (Figure 3b), indicating a poleward expansion of the equatorward meridional wind associated with the Hadley cell. A similar pattern emerges in the zonal-mean vertical pressure velocity field (not shown), with the region of maximum descent in the subtropics shifting poleward. Accordingly, the subtropical minimum in total precipitation shifts poleward in the 4xCO2+4K case (Figure 3c). Without the 4 K increase in SSTs, a widening of the Hadley cell is not seen in the zonal-mean meridional wind (Figure 3a) or precipitation rate (Figure 3c) fields for the 4xCO2 simulation. The tropical-average (15°S–15°N) precipitation rate increases by only 0.58% K^{-1} from the 1xCO2 simulation to the 4xCO2+4K simulation despite the increased SSTs, perhaps because mean upwelling decreases and static stability (not shown) increases between 10°S and 10°N (Fu & Wu, 2024). However, the global-mean precipitation rate increases by 1.86% K^{-1}, which closely matches the expected 2%–3% K^{-1} change derived from thermodynamic arguments (Held & Soden, 2006; Jeevanjee & Romps, 2018) and confirmed by comprehensive climate models (Richter et al., 2022). Like the poleward shift of the tropospheric mid-latitude zonal jet, the expansion of the Hadley cell is frequently observed in comprehensive climate models (Medeiros et al., 2015), and lends credibility to these idealized simulations.

3.2. Stratospheric Mean Circulation Response

Increasing atmospheric carbon dioxide also impacts the dynamics of the stratosphere, although the strength of the modeled circulation depends on the SSTs. The strength of the BDC, represented by the residual stream function, and its principal driver, the Eliassen-Palm flux divergence body force, are displayed in subplots (c,d) and (a,b), respectively, of Figure 4. As before, the black contours represent the control simulation climatology and the shaded contours quantify the difference between the elevated CO_2 and control simulations. Despite the lack of topography in the aqua-planet model, the upwelling mass flux at 68 hPa in the control simulation is 5.1×10^9 kg s^{-1}, which is very similar to the annual average 70 hPa upwelling mass flux in multiple reanalysis data products (Abalos et al., 2021). This suggests that forcing from Rossby waves generated by baroclinic instability in the troposphere compensates for the lack of stationary orographic Rossby waves in the aqua-planet model (Eichelberger & Hartmann, 2005). Residual stream function anomalies suggestive of a strengthened BDC are about an order of magnitude larger in the 4xCO2+4K simulation (Figure 4d) than the 4xCO2 simulation (Figure 4c). This is likely because increased SSTs in the 4xCO2+4K case strengthen resolved wave driving in both hemispheres of the lower stratosphere between the equator and 50°N/S (Figure 4b), whereas resolved wave driving is relatively unchanged in the 4xCO2 case (Figure 4a).

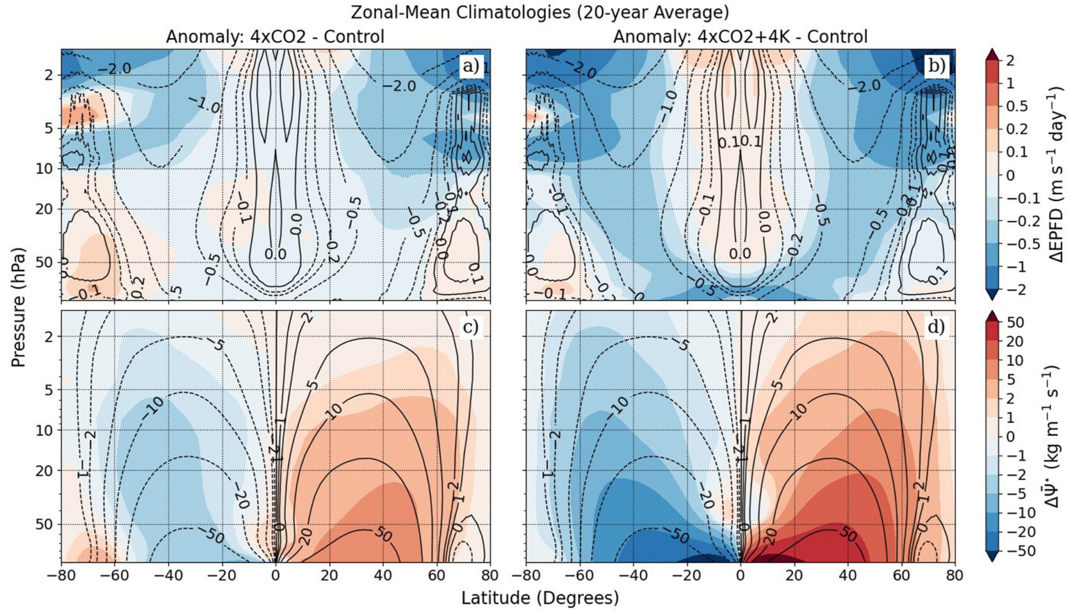

Figure 4. Zonal-mean differences between the 20-year control and elevated CO_2 stratospheric climatologies. Shaded contours in (a) and (b) display the Eliassen-Palm flux divergence body force (EPFD; Equation 11) differences between the control and elevated CO_2 simulations, while the black contours are the control EPFD body force. (c) and (d) are the same, but for differences in the residual stream function, defined by Equation 4.

The response of the stratospheric zonal winds also differs between the 4xCO2 and 4xCO2+4K simulations. The 4xCO2 simulation exhibits opposing anomalies in stratospheric zonal wind poleward of 50°N/S (Figure 2a). Given the hemispheric symmetry of the aqua-planet configuration, these zonal wind anomalies are likely insignificant. Thus, the impact of stratospheric cooling of up to 20 K associated with increased long-wave emission from CO_2 (Figure 2c) on the strength of the aqua-planet polar vortex appears to be minimal. On the other hand, there is a pronounced deceleration of the polar vortex in both hemispheres of the 4xCO2+4K stratosphere above 5 hPa (Figure 2b). This weakening may be a result of the increased planetary-scale wave drag exerted on the zonal winds in the upper polar stratosphere (Figure 4b), which also locally reduces the large-scale radiative cooling (Figure 2d) compared to the 4xCO2 simulation. Below this level, hemispheric asymmetries in the 4xCO2+4K polar stratospheric zonal wind anomalies hinder the identification of a robust polar vortex response to surface warming. The asymmetry between the Northern and Southern Hemisphere zonal wind responses to idealized climate change may be due to sudden stratospheric warming events caused by Rossby wave breaking. During such an event, the stratospheric zonal jet reverses direction, and the accumulated effect of many events is noise in the zonal wind field. In the aqua-planet simulation, SSWs were identified when the 10 hPa zonal wind at 60°N/S dropped below zero. For the control case, this occurred five times in the Northern Hemisphere and 14 times in the Southern Hemisphere. A longer simulation would likely be needed to minimize the effects of this internal variability on the zonal wind climatology.

Although the 4xCO2 and 4xCO2+4K simulations both exhibit the expected stratospheric cooling in response to elevated CO_2, the tropospheric response to climate change is amplified when SSTs are warmed in the 4xCO2+4K experiment. The difference in climate response between the 4xCO2 and 4xCO2+4K simulations confirms the importance of surface warming as opposed to rapid thermal adjustments from the CO_2 radiative forcing in shaping the tropospheric general circulation in a changing climate. In agreement with these results, Ceppi et al. (2018) found that the temperature response to CO_2 direct radiative forcing in an atmosphere-only configuration of version 4 of CAM (CAM4) was limited to stratospheric cooling when SSTs were fixed. Similarly, Grise and Polvani (2014) determined that the majority of the poleward shift in the Southern Hemisphere tropospheric zonal jet in fully coupled CMIP5 models can be attributed to the indirect impact of warming SSTs. Thus, despite the simplicity of the aqua-planet configuration, the response of the aqua-planet mean circulation to increased CO_2 and higher SSTs is consistent with comprehensive climate models.

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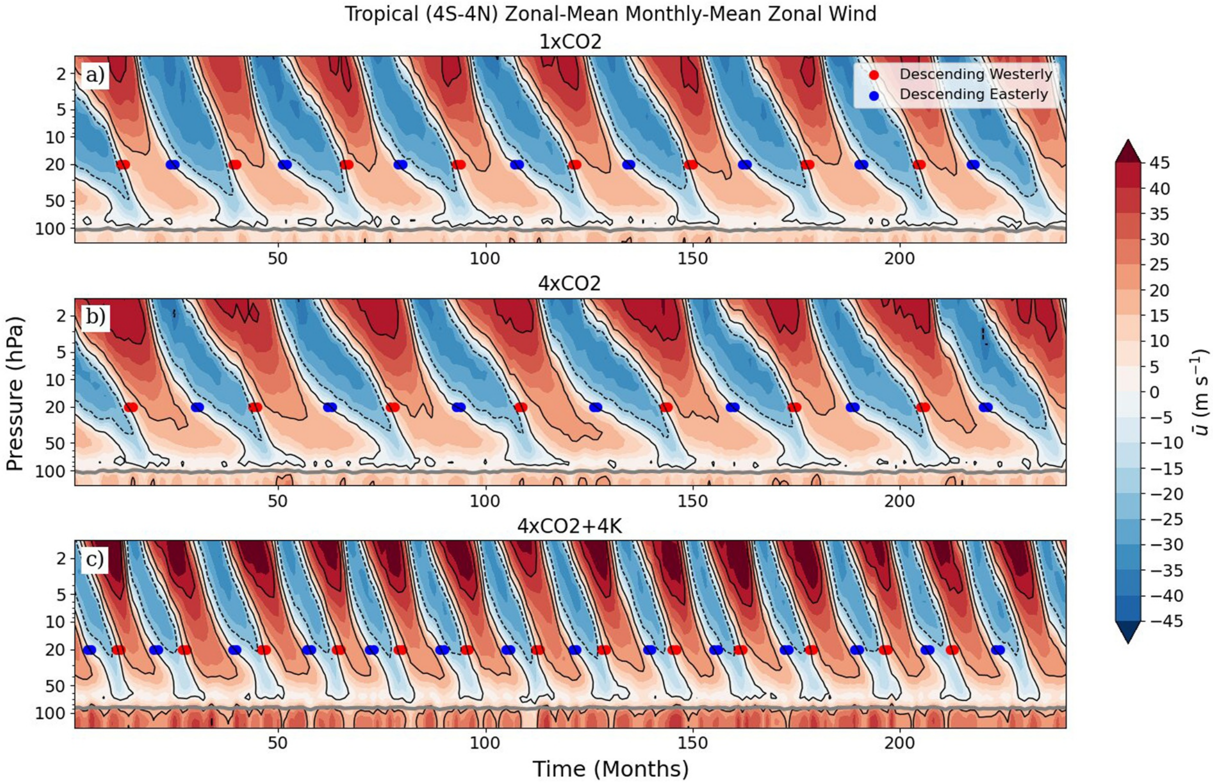

Figure 5. Zonal-mean, monthly mean zonal wind in the stratosphere averaged over 4°S-4°N for (a) the control simulation, (b) 4xCO2, and (c) 4xCO2+4K. Shaded contours represent zonal wind values which were sampled every month in the model. The thick gray line is the tropopause. Red dots represent descending westerly months, and blue dots represent descending easterly months.

3.3. Momentum Budget Analysis of QBO Changes

The QBO in the aqua-planet configuration of CAM7 is sensitive to the concentration of carbon dioxide, especially if surface warming is accounted for. In this section, the drivers of the changing QBO will be analyzed using the TEM framework detailed in Section 2.3.2. To reiterate, the oscillation of the tropical stratospheric zonal winds is affected by advection due to the residual (Brewer-Dobson) circulation, resolved and parameterized wave driving, and numerical diffusion (Equation 7). Changes in these driving terms modulate the QBO period by altering the rate of descent of the QBO jets.

Figure 5 provides a visual summary of the characteristics of the QBO in the control and elevated carbon dioxide simulations. In the lower stratosphere (below 10 hPa) of the control (1xCO2) simulation (Figure 5a), the QBO easterlies are stronger than the QBO westerlies, in agreement with the observed QBO (Naujokat, 1986). In all cases, the amplitude of the QBO continues to increase in the vertical direction because the aqua-planet configuration lacks the observed semi-annual oscillation (SAO) in the upper stratosphere above 5 hPa. Normally, resolved and parameterized waves that are not absorbed by the QBO deposit their momentum in the SAO region (Richter & Garcia, 2006). However, unlike the QBO, the SAO depends on the seasonal cycle and cannot be simulated in the aqua-planet model. As a result, the vertical extent of the QBO is limited only by a sponge layer diffusion process near the model top. The QBO behaves similarly in other idealized models that do not simulate the SAO (e.g., Garfinkel et al., 2022). Figure 6 displays the latitudinal structure of the aqua-planet QBO in the descending westerly (a–c) and descending easterly (d–f) phases for each experiment. The black contours are the composite zonal winds, the shaded contours are composite temperature anomalies with respect to the 20-year climatology, and the arrows represent composite anomalies in the meridional residual circulation. The aqua-planet residual circulation anomalies reveal a realistic QBO secondary circulation, which is characterized by descending motion in the westerly shear zone (Figures 6a–6c), ascending motion in the easterly shear zone (Figures 6d–6f), and oppositely directed motion in the subtropics to close the meridional cells (Pahlavan, Wallace, et al., 2021). As expected, the temperature anomalies maintain the thermal wind balance within the QBO

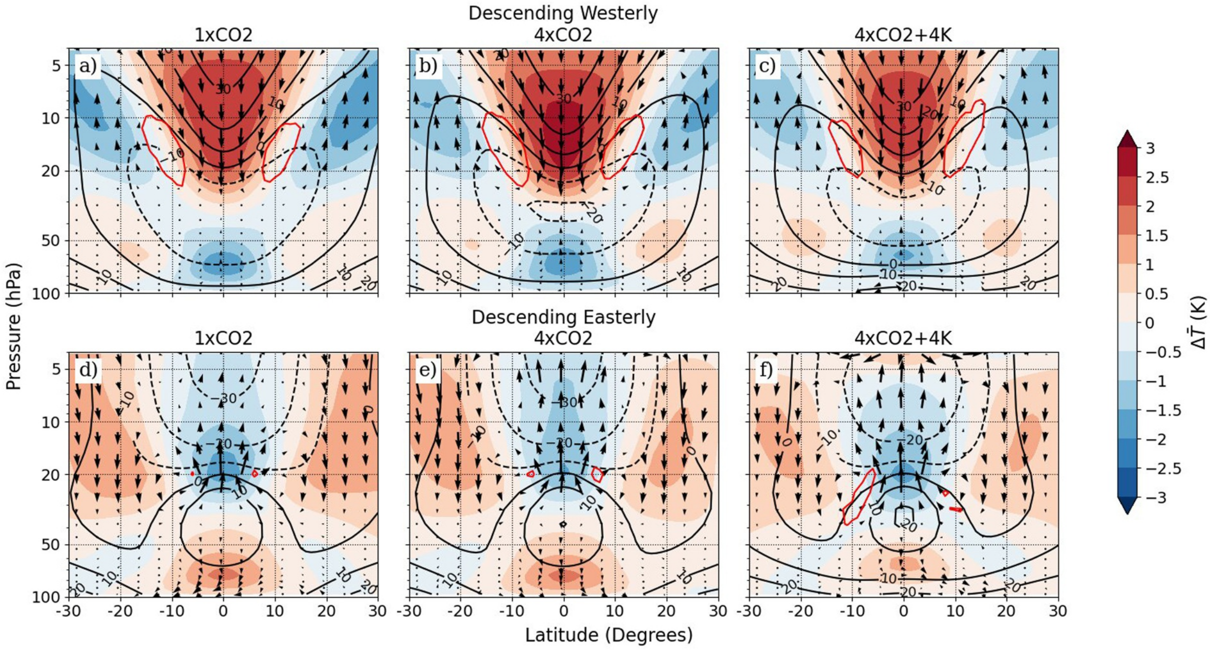

Figure 6. Descending westerly (a–c) and descending easterly (d–f) zonal mean composites of the Quasi-Biennial Oscillation secondary circulation. Black contours represent composite zonal winds, and shaded contours are composite temperature anomalies with respect to the 20-year climatology. The arrows are composite anomalies in the residual meridional circulation, $(\bar{v}^*, \bar{w}^*)$, where the vertical residual velocity has been scaled according to Equation 6. The maximum horizontal component of the descending westerly (a–c) residual circulation arrows is 0.12 m s^{-1}, and the longest vertical component is 0.44 mm s^{-1}. In the descending easterly phase (d–f) the arrows have been scaled to be twice as long as in the descending westerly phase, and the longest vertical and horizontal components are 0.17 m s^{-1} and 0.27 mm s^{-1}, respectively. Areas inside the red contour satisfy the necessary condition for barotropic instability, $\beta - \bar{u}_{yy} < 0$, where β is the meridional derivative of the Coriolis parameter.

secondary circulation (Anstey et al., 2022). There is a clear decrease in the period of the 4xCO2+4K QBO relative to the control simulation (Figure 5c), and the 4xCO2 period lengthens slightly (Figure 5b). The amplitude of the QBO also decreases noticeably in the lower stratosphere for the 4xCO2+4K case in response to the combined influence of tropospheric expansion and stronger upwelling. The 20 hPa QBO amplitudes and periods calculated with the transition times method described in Section 2.3.1 are listed in Table 1 for all experiments.

These amplitude and period changes can be explained in terms of the TEM momentum budget, as defined in Equation 7. The terms in the momentum equation are displayed in Figure 7 for the descending westerly and easterly phases of the QBO. The vertical tendency profiles are composites averaged over the transition times of the QBO, as discussed in Section 2.3.1. The difference profiles in panels b, c, e, and f are obtained by subtracting the control tendency profiles (a,d) from the 4xCO2 and 4xCO2+4K profiles. Figure 7a suggests that the positive zonal wind tendency in the descending westerly shear zone of the control simulation is mostly due to resolved wave forcing (EPFD), with a smaller positive contribution from parameterized gravity waves (GWD) and a negative contribution from advection by the residual circulation (ADV), which always acts against the descent of the QBO jets. The advection tendency is much smaller in the westerly shear zone (Figure 7a) than the easterly shear zone (Figure 7d) because the QBO secondary circulation introduces downward motion below the descending westerly jet (Figures 6a–6c). The resolved wave tendency in the control run near the westerly shear zone is 0.41 m s^{-1} day^{-1}, about three times larger than the gravity wave tendency, 0.13 m s^{-1} day^{-1}. This ratio disagrees with the consensus of GCMs that resolved and parameterized waves each make up about half of the total westerly tendency (Giorgetta et al., 2006; Holt et al., 2022; Krismer & Giorgetta, 2014). However, this lack of westerly gravity wave tendency in the aqua-planet model can be explained by critical level filtering in the anomalous tropical upper tropospheric zonal jet near 150 hPa (Figure 2b). The control momentum budget at 20 hPa during the descending easterly phase of the QBO is shown in Figure 7d. Unlike the

Table 1
Number of Full Quasi-Biennial Oscillation (QBO) Cycles, QBO Amplitude, QBO Period, and Their Standard Deviations at 20 hPa Determined by the Transition Times Method for Each Experiment

	N_{cycles}	Amplitude (m s^{-1})	Period (months)
1xCO2	8	23.8 ± 0.7	27.6 ± 0.7
4xCO2	7	24.0 ± 0.6	31.6 ± 1.8
4xCO2+4K	13	21.6 ± 0.9	16.9 ± 1.1

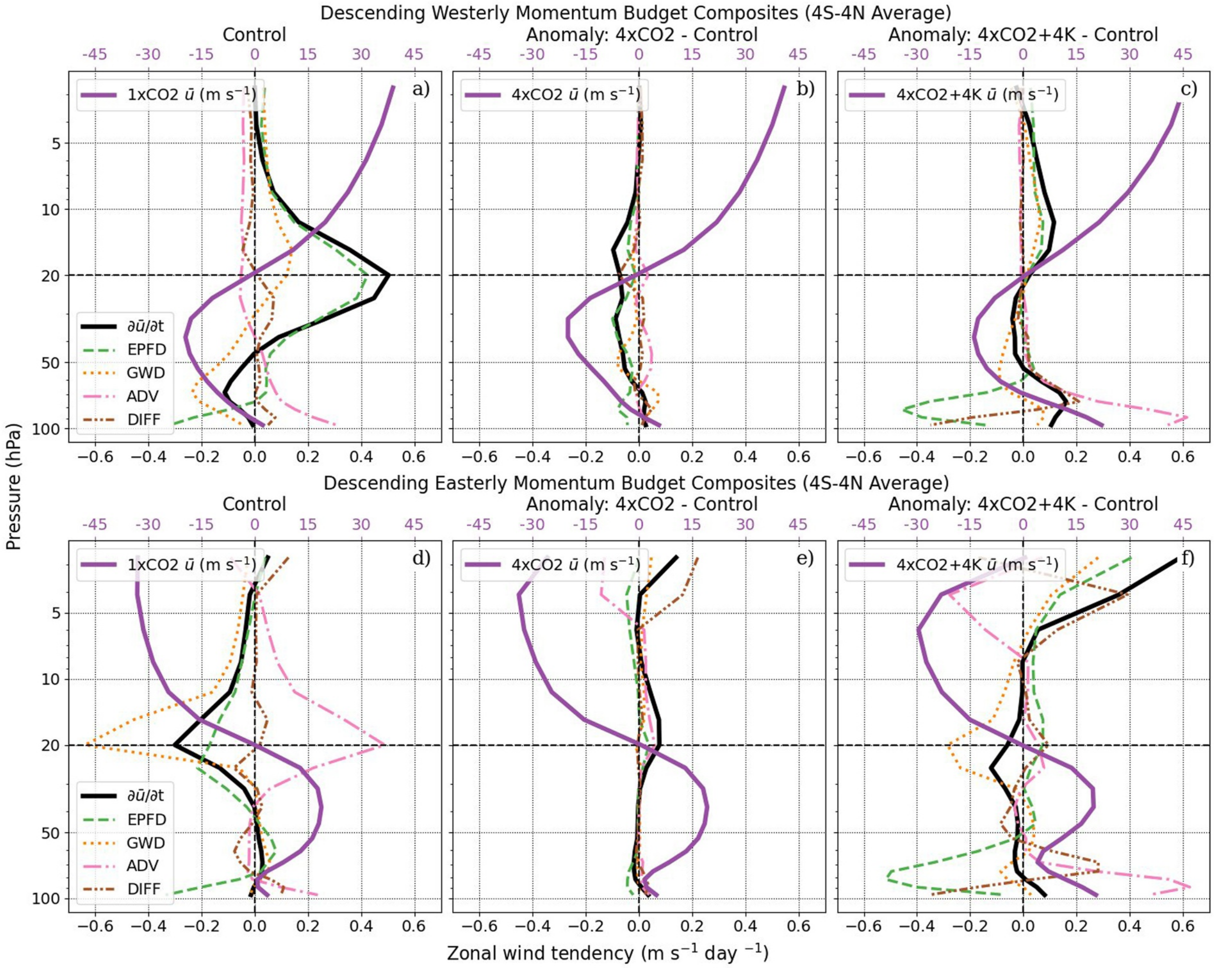

Figure 7. Composites of zonal wind tendency due to the Transformed Eulerian Mean zonal momentum budget terms (Equation 7) for times when the zonal-mean zonal wind is westerly above 20 hPa (descending westerly; a–c) and easterly above 20 hPa (descending easterly; d–f). $\partial \bar{u}/\partial t$ is the total zonal wind time tendency, EPFD is the zonal wind time tendency due to EP Flux divergence, GWD is the zonal wind time tendency due to gravity wave drag, ADV is the zonal wind time tendency due to advection by the residual circulation, and DIFF is the diffusion zonal wind time tendency. The right two panels in each row represent anomalies in the 4xCO2 and 4xCO2+4K momentum budget terms with respect to the control simulation. In all panels, the dark purple profile is the background zonal wind for the given simulation. All composites were averaged from 4°S–4°N because the meridional structure of the EPFD changes outside of this range.

westerly phase, the descending easterly QBO jet is primarily driven by gravity wave drag. The gravity wave tendency is about -0.62 m s^{-1} day^{-1}, while the resolved wave forcing is only -0.17 m s^{-1} day^{-1} at 20 hPa. Furthermore, the descent of the easterly jet is much more strongly opposed by the residual circulation (ADV) than the descent of the westerly jet.

In both phases, the numerical diffusion tendency is zero at the shear zone but non-zero above and below, rendering its impact on the QBO period uncertain. A very similar tendency pattern has been observed in simulations with idealized (Yao & Jablonowski, 2013, 2015) and comprehensive (Kim & Chun, 2015a) model physics, and may be caused by the fourth-order hyper-diffusion scheme used in the SE dynamical core. Although the SE hyper-diffusion operates horizontally (Lauritzen et al., 2018), by reducing meridional gradients in zonal wind poleward of the equatorial shear zones (Figure 6) it could decrease the strength of the descending jet and increase the strength of the jet being replaced as observed in Figures 7a and 7d. The resultant smoothing of the vertical gradient in zonal wind is reminiscent of vertical eddy diffusion, as noted by Kim and Chun (2015a). Since the diffusion tendency counteracts the total zonal wind tendency above the shear line and supplements it below the shear line in both the descending westerly (Figure 7a) and descending easterly (Figure 7d) phases, it is unclear whether numerical diffusion accelerates or decelerates the descent of the QBO, although its effects are overshadowed by resolved and parameterized wave forcing.

3.3.1. Resolved Wave Forcing

Changes in resolved wave forcing increase the 4xCO2 QBO period, speed up the descent of the 4xCO2+4K westerly jet, and slow the descent of the 4xCO2+4K easterly jet. Resolved wave forcing (EPFD) makes up the majority of the negative zonal wind tendency anomaly below 20 hPa in the descending westerly phase of the 4xCO2 QBO (Figure 7b). This negative anomaly in the zonal wind tendency delays the downward propagation of the QBO westerly jet, lengthening the 4xCO2 QBO period. On the contrary, increased resolved wave forcing strengthens the westerly zonal wind tendency between 5 and 20 hPa in the descending westerly phase of the 4xCO2+4K QBO (Figure 7c). Since this positive zonal wind tendency anomaly is located below the peak of the QBO westerlies, it accelerates the descending westerlies and decreases the 4xCO2+4K period. In the descending easterly phase of the 4xCO2+4K QBO, an increase in the resolved wave drag of about 0.07 m s^{-1} day^{-1} at 20 hPa (Figure 7f) partially counteracts the downward acceleration of the easterly jet by increased parameterized gravity wave drag. Acceleration of the descending westerlies and deceleration of the descending easterlies by resolved waves is also observed in climate change simulations using a fully coupled configuration of CAM (Lee et al., 2024). Even larger resolved wave forcing anomalies occur in both phases of the 4xCO2+4K QBO near 80 hPa (Figures 7c and 7f), and are suggestive of an upward shift of the existing control profile (Figures 7a and 7d) associated with tropospheric expansion. Given their alignment with a minimum in zonal wind magnitude, these resolved wave forcing tendencies can be attributed to the dissipation of equatorial waves with low zonal phase speeds ($|c| \leq 10$ m s^{-1}) (Kawatani et al., 2009; Krismer & Giorgetta, 2014). Since this negative, lower-stratospheric resolved wave forcing is nearly balanced by a positive tendency due to vertical advection, the net zonal wind forcing is negligible.

The changes in resolved wave forcing for the 4xCO2 and 4xCO2+4K simulations can be attributed to different types of equatorial waves. Figure 8 displays a wave-by-wave decomposition of the control (a,d) and anomaly (b,c,e,f) resolved wave forcing for the descending westerly and descending easterly QBO phases. For the descending westerly phase of the control simulation (Figure 8a), the positive resolved wave tendency is largely due to Kelvin waves with a smaller positive contribution from IG and resolved SSG waves and a negative contribution from MRG waves, in agreement with observations (Pahlavan, Fu, et al., 2021) and climate model results (Kim & Chun, 2015a). The decrease in resolved wave driving of the descending westerly phase in the 4xCO2 simulation is almost entirely due to a decrease in Kelvin wave forcing (Figure 8b). The positive trend in resolved wave driving of the descending westerlies for 4xCO2+4K is due to a combination of increased Kelvin and IG wave forcing (Figure 8c), which is consistent with comprehensive climate model experiments (Lee et al., 2024). The negative resolved wave tendency in the descending easterly shear zone of the control simulation (Figure 8d) is about half the magnitude of the corresponding descending westerly tendency (Figure 8a) and is attributed to MRG, IG, and SSG wave forcing. Reanalysis and comprehensive climate model experiments generally agree with this result, although MRG waves tend to be under-represented by comprehensive climate models, including CESM (Holt et al., 2022), and aqua-planet models in general (Rios-Berrios et al., 2022).

That an appreciable MRG forcing still exists in the descending easterly shear zone suggests that the stratospheric MRG waves originate locally and do not propagate upward from the troposphere. It has been shown in studies of reanalysis (Pahlavan, Fu, et al., 2021; Mahó et al., 2025) and climate model (Garcia & Richter, 2019; Kim & Chun, 2015a) data that barotropic instability near the flanks of the westerly QBO jet can generate MRG waves. Indeed, Figures 6a–6c reveal that the necessary condition for barotropic instability, $(\beta - \bar{u}_{yy}) < 0$ (Simpson et al., 2025), is satisfied near 10°N/S in the descending westerly phase of the QBO. Furthermore, it was verified that the MRG-filtered Eliassen-Palm flux vectors (Equations 10a and 10b) indicate meridional, rather than upward, wave propagation. While there is little change in the resolved wave drag for the 4xCO2 descending easterly phase (Figure 8e), the decreased resolved wave driving of the 4xCO2+4K descending easterlies (Figure 8f) is a result of weakened MRG, ER, and IG zonal wind tendencies. A decrease in MRG wave forcing of the descending easterlies in a warmer climate was also noted by Lee et al. (2024), who suggested that this wave generation mechanism will weaken due to the decrease in QBO amplitude. It is possible that other resolved wave forcing anomalies shown in Figure 8 are a response to existing zonal wind anomalies, since the Eliassen-Palm flux (Equations 10a and 10b) depends on the background zonal wind, $\bar{u}$. To clarify the source of the resolved wave forcing anomalies, the response of tropospheric equatorial waves to increased SSTs and CO_2 concentrations is analyzed next.

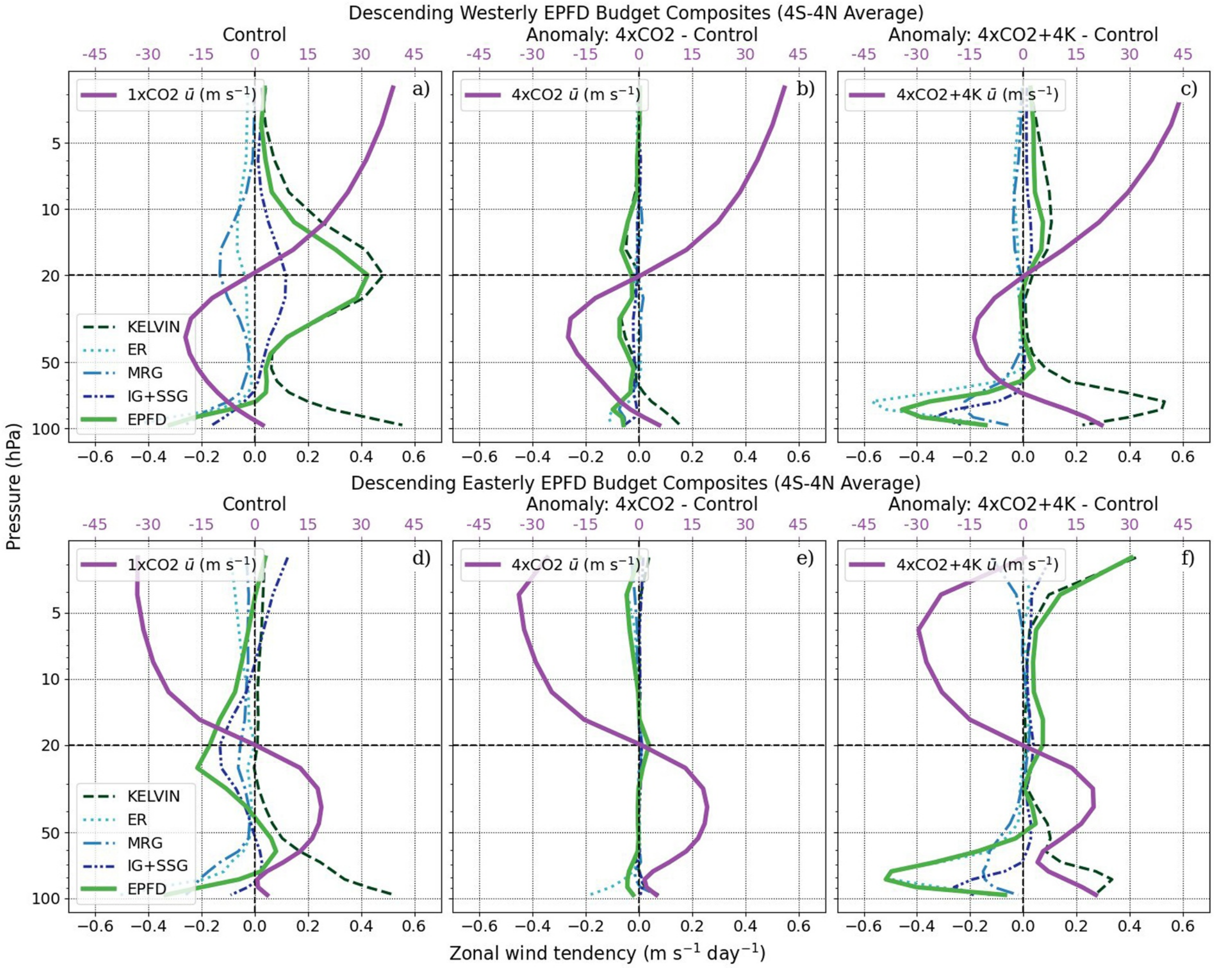

Figure 8. As in Figure 7 but for the resolved wave components of the Eliassen-Palm flux divergence body force (EPFD).

Convectively coupled Kelvin wave activity increases in the 4xCO2+4K simulation, directly impacting the QBO period and indirectly modulating the flux of parameterized gravity waves in the stratosphere. Figure 9 contains the symmetric (a,b,c) and antisymmetric (d,e,f) power spectra of the convective precipitation rate averaged over the tropical region (15°S-15°N) and calculated following the method of Wheeler and Kiladis (1999) described in Section 2.3.3. In agreement with other aqua-planet simulations (Nakajima et al., 2013; Rios-Berrios et al., 2020, 2022), the precipitation spectrum is dominated by Kelvin waves in the symmetric component and advective disturbances in the antisymmetric component. The aqua-planet underestimates MRG, $n = 0$ IG, and $n = 1$ IG power compared to comprehensive models and observations (Holt et al., 2022). Convectively coupled Kelvin wave power increases for the 4xCO2+4K simulation (Figure 9c) and shifts to larger phase speeds relative to the control simulation (Figure 9a), since the band of significant Kelvin wave power lies along dispersion curves with a larger equivalent depth. These trends are also seen in aqua-planet (Chien & Kim, 2024) and fully coupled CMIP6 (Bartana et al., 2023) simulations of a warmer climate. Increased Kelvin wave activity strengthens the anomalous tropical upper tropospheric zonal jet by means of critical level absorption, indicated by increased resolved wave drag (EPFD) just below the core of the tropical zonal jet in Figure 10, which displays climatological profiles of zonal-mean, tropical-average zonal wind (a), resolved wave forcing (a), and gravity wave momentum flux (b) for all simulations. Amplified 4xCO2+4K Kelvin wave activity also explains the increase in westerly QBO Kelvin wave forcing shown in Figure 8c that contributes to a shorter QBO period. No such change in Kelvin wave activity is seen in the symmetric power spectrum for the 4xCO2 simulation (Figure 9b), suggesting that SSTs are an important driver of large-scale wave activity in an aqua-planet. Similarly, little change in antisymmetric power relative to the control

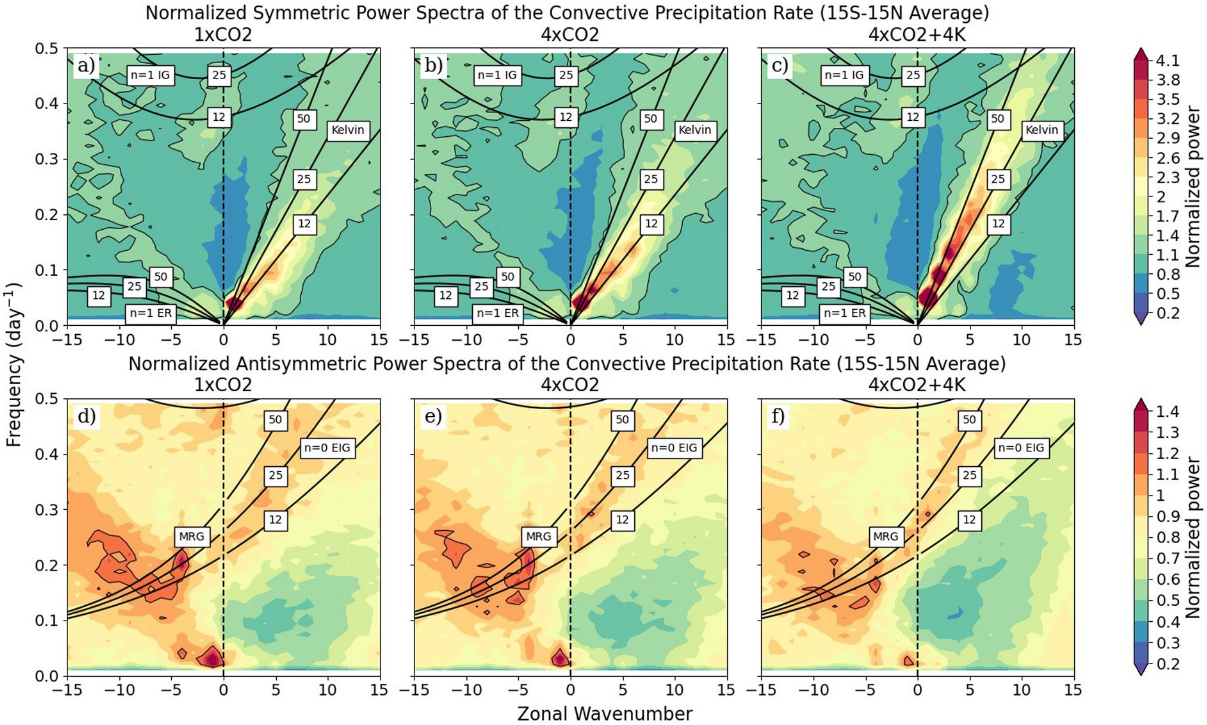

Figure 9. Wavenumber-frequency plots of symmetric (top row) and antisymmetric (bottom row) convective precipitation power for (a, d) the control simulation, (b, e) 4xCO2, and (c, f) 4xCO2+4K following the method of Wheeler and Kiladis (1999). The symmetric power was normalized using a smoothed background spectrum and was averaged from 15°S–15°N. The shallow water dispersion curves are shown in black for relevant equatorial wave types, which are labeled in the boxes. Each boxed number is the equivalent depth in meters of a shallow water system that would be needed to produce such a wave.

(Figure 9d) simulation is observed for the 4xCO2 (Figure 9e) case. A small decrease in significant antisymmetric power is present in the MRG wavenumber-frequency range for the 4xCO2+4K (Figure 9f) simulation, but it is not possible to disentangle this trend from changes in AD activity. However, given that a large portion of the MRG-related QBO forcing originates from MRG waves generated in the stratosphere (Pahlavan, Fu,

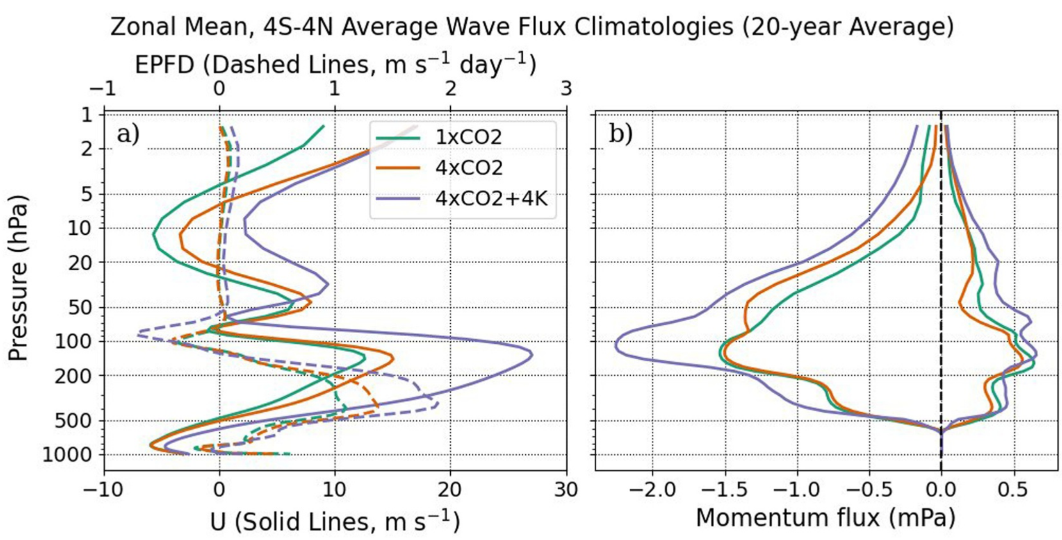

Figure 10. Zonal-mean, 20-year climatologies of (a) EPFD body force (dashed) and zonal wind (solid) and (b) gravity wave momentum flux averaged from 4°S to 4°N for all three simulations as noted in the legend.

et al., 2021; Mahó et al., 2025), it is unlikely that small trends in convectively coupled MRG activity would impact the QBO.

3.3.2. Parameterized Wave Forcing

Parameterized gravity wave forcing is the primary driver of the shorter 4xCO2+4K QBO period. In the descending westerly phase of the 4xCO2+4K QBO, the zonal wind tendency due to gravity wave drag increases by as much as 0.07 m s^{-1} day^{-1} above 20 hPa and below the core of the westerly QBO jet (Figure 7c), accelerating its descent. Similarly, in the descending easterly phase the gravity wave drag increases in magnitude by 0.29 m s^{-1} day^{-1} near the 20 hPa shear zone, hastening the descent of the QBO easterlies (Figure 7f). There is little change in gravity wave driving in the 4xCO2 simulation because the precipitation (Figure 3c) and mean tropospheric state (Figures 2a and 2c) are largely insensitive to the quadrupling of atmospheric carbon dioxide. Parameterized gravity waves are a major source of uncertainty in model projections of the QBO period, but the decreased QBO period observed in the 4xCO2+4K simulation is at least consistent with the behavior of the QBO for fully coupled configurations of CESM (Lee et al., 2024; Richter et al., 2022).

Increases in stratospheric gravity wave momentum flux in the 4xCO2+4K simulation are partly due to intensified deep convection. The Beres et al. (2005) convective gravity wave scheme utilized by CAM distributes momentum flux across a range of gravity wave phase speeds according to the depth of the convective heating region, the mean wind within the convective heating region, and the maximum convective heating rate. The mean wind in the convective cell strongly influences the symmetry of the gravity wave phase speed spectrum and tends to increase the amplitude of momentum flux in the opposite direction (Lee et al., 2024; Liu et al., 2021). The heating depth and maximum heating rate influence the spectral distribution and amplitude, respectively, of the parameterized gravity waves. For "very strong" convective events, defined as grid points with maximum heating rates greater than or equal to the 99th percentile (Franke et al., 2023), the tropical-average (15°S–15°N) heating rate increased by 24.5% between the control and 4xCO2+4K simulations, consistent with comprehensive model results (Franke et al., 2023; Lee et al., 2024). Again averaging over very strong convective events, the tropical average heating depth was found to increase by 9.9% as a result of SST warming. A similar relative change in tropical convection depth was observed in storm-resolving simulations subject to abrupt increases in SSTs (Franke et al., 2023). The enhanced 4xCO2+4K westerly (positive) gravity wave momentum flux evident in Figure 10b is likely an effect of the increased tropical convective heating rate, since the gravity wave momentum flux is proportional to the square of the maximum heating rate in the Beres et al. (2005) scheme. This increase in westerly gravity wave momentum flux ultimately accelerates the descent of the 4xCO2+4K westerly QBO jet (Figure 7c). Thus, although the tropical convective precipitation rate (Figure 3c) is relatively insensitive to warming SSTs, the aqua-planet deep convection intensifies and drives greater gravity wave forcing of the QBO.

However, much of the change in easterly gravity wave momentum flux in the 4xCO2+4K simulation is caused by trends in the tropical zonal wind profile. Specifically, the strengthened tropical upper tropospheric zonal jet near 150 hPa (Figure 2b) increases the flux of easterly gravity waves by introducing asymmetry in the source gravity wave spectrum. A strong tropical upper tropospheric zonal jet is a peculiar characteristic of aqua-planet simulations, and is exhibited by many of the aqua-planet models that participated in the APE project (Williamson et al., 2012). Comprehensive climate models do not exhibit a change in the tropical zonal wind profile in a warmer climate (Richter et al., 2022). Instead, as discussed above, changes in gravity wave momentum flux are connected to changes in tropical convection (Richter et al., 2022). However, Lee et al. (2024) note that the increase in gravity wave drag in a future climate obeying the Shared Socioeconomic Pathway 3–7.0 (7.0 W m^{-2} radiative forcing by 2100) scenario is only significant during the descending easterly phase of the QBO. They find that an increase in westerly zonal winds in the upper troposphere and lower stratosphere during the descending easterly phase causes the Beres et al. (2005) scheme to amplify the amount of easterly momentum flux entering the stratosphere. It is thus plausible that the tropical zonal wind profile trends (Figure 10a) drive a large increase in easterly (negative) momentum flux in the stratosphere (Figure 10b), especially since the increase in upper tropospheric zonal winds in the 4xCO2+4K simulation (15 m s^{-1}) is much greater than the zonal wind anomalies in the fully coupled simulation (2.5 m s^{-1}; Lee et al., 2024).

3.3.3. Residual Circulation Forcing

The strengthened BDC and associated tropical upwelling lengthen the 4xCO2 QBO period. In the descending easterly phase of the 4xCO2 QBO, an increase in the zonal wind tendency due to advection by the residual circulation comprises most of the total positive zonal wind anomaly of about 0.08 m s^{-1} day^{-1} (Figure 7e) which weakens the net negative zonal wind tendency in the easterly shear zone and slows the descent of the QBO easterly jet. A similar positive zonal wind tendency anomaly due to advection appears in the descending easterly phase of the 4xCO2+4K QBO (Figure 7f), but is overruled by the large increase in the magnitude of gravity wave drag. The advection term has a much larger relative impact on the 4xCO2 QBO period because in that simulation there is not a concurrent increase in gravity wave momentum flux to counteract it. Indeed, results from the QBOi (Richter et al., 2022) suggest that in full-complexity climate models without a flow-dependent non-orographic gravity wave scheme, such as the Max Planck Institute ECHAM5sh or UK Met Office UMGA7 models, increased tropical upwelling dominates other QBO drivers to yield a future QBO with a longer period.

The weakened QBO amplitude in the 4xCO2+4K experiment is likely a combined result of the strengthened residual upwelling and tropospheric expansion in the aqua-planet tropics. Climatological, tropical-average profiles of vertical residual velocity (Equation 6) and QBO amplitude are compiled for all simulations in Figure 11. The amplitude of the 4xCO2+4K QBO decreases by as much as 64% in the lower stratosphere (Figure 11b), consistent with the 35% increase in tropical-average (20°S–20°N) upwelling in the same region (Figure 11a), which falls in the middle of the 15%–75% change in vertical velocity observed in the QBOi models (Richter et al., 2022). The definition of the tropics has been expanded here to include the entire upwelling branch of the BDC. A weakening of the QBO in response to greater upwelling is robustly predicted by comprehensive climate models (Richter et al., 2022), and is consistent with theoretical studies (Match & Fueglistaler, 2020; Saravanan, 1990) connecting the depth of the QBO buffer zone with tropical upwelling. However, the minima in the 4xCO2+4K vertical residual velocity (Figure 11a) and QBO amplitude (Figure 11b) occur about 15 hPa higher than in the control simulation, in tandem with the elevation of the tropical tropopause by roughly 15 hPa. This suggests that the changes in the 4xCO2+4K QBO amplitude and vertical residual velocity could be explained by an upwards shift in the model dynamics. Indeed, Oberländer-Hayn et al. (2016) argued based on chemistry climate model results that the future BDC trends encapsulate an upward shift, rather than a strengthening, of the residual circulation. Furthermore, Match and Fueglistaler (2021) proposed that both the increased upwelling and decreased QBO amplitude in the lower stratosphere are a result of tropospheric expansion, which effectively shifts temperatures and their associated dynamical responses upwards. Tropospheric expansion is a well-understood thermodynamical response to climate change that is agreed upon by climate models, and may be why QBO amplitude changes are remarkably uniform across models. Since Figure 11a also suggests a translation of the upwelling climatology to higher vertical residual velocities, it is possible that the aqua-planet BDC trends are a combination of strengthening and vertical expansion. Due to the confounding influence of tropospheric expansion, it is difficult to extricate the effect of upwelling on the aqua-planet QBO amplitude.

4. Discussion and Conclusions

In this study, the response of the QBO to increasing greenhouse gas concentrations was analyzed in an idealized aqua-planet configuration of CAM7. Despite the removal of external sources of variability like the seasonal cycle and ENSO, lack of topography, and the prescription of zonally symmetric SSTs, a surprisingly realistic QBO was simulated. Although the aqua-planet lacks an SAO in the upper stratosphere, and thus simulates unrealistically large QBO amplitudes in that region, the response of the QBO to climate forcing in the middle and lower stratosphere was found to be quite similar to that observed in comprehensive climate models. To evaluate the sensitivity of the aqua-planet QBO to climate change, two idealized climate change experiments were conducted. The first was identical to the control simulation except for the quadrupling of the constant atmospheric carbon dioxide mixing ratio. The purpose of this experiment was to diagnose the short-timescale radiative impact of the carbon dioxide without considering surface warming. In the second experiment, carbon dioxide was quadrupled and SSTs were raised uniformly by 4 K to approximate the ocean response to increasing greenhouse gas

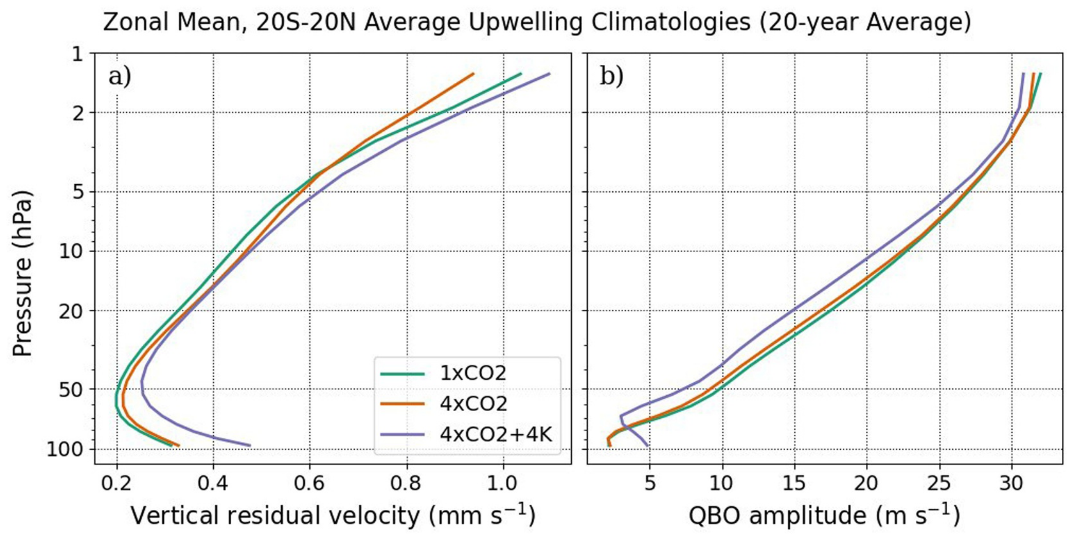

Figure 11. Zonal-mean, 20-year climatologies of (a) vertical residual velocity and (b) Quasi-Biennial Oscillation amplitude calculated using the Dunkerton and Delisi (1985) method averaged from 20°S to 20°N for all three simulations as noted in the legend.

concentrations. Increased carbon dioxide concentrations caused stratospheric cooling in both cases, but changes to the tropospheric mean circulation and the wave-driven BDC were much larger when the SSTs were increased.

The aqua-planet QBO was found to be minimally sensitive to the carbon dioxide radiative forcing, but was strongly impacted by increased SSTs. There was only a small increase in the QBO period observed in the 4xCO2 simulation with respect to the control case, mostly driven by strengthened tropical upwelling. In contrast, in the 4xCO2+4K simulation the increased SSTs led to increases in Kelvin wave driving of the descending QBO westerlies, gravity wave driving of both phases of the QBO, and planetary-scale wave drag in the lower subtropical stratosphere. The net effects of this increased wave drag were a shorter QBO period and a stronger BDC. A decrease in the 4xCO2+4K QBO amplitude near 70 hPa consistent with comprehensive model results (e.g., Butchart et al., 2020; Lee et al., 2024; Richter et al., 2022) was associated with changes in tropical upwelling, but may also be explained by an upward shift of atmospheric dynamics due to tropospheric expansion (Match & Fueglistaler, 2021). In the 4xCO2+4K simulation, stratospheric Kelvin wave anomalies were connected to changes in the tropical precipitation spectrum, and gravity wave anomalies were linked to intensified deep convection and modified background zonal winds. These results suggest that the QBO is more responsive to surface warming than the rapid adjustment of stratospheric temperatures caused by increased carbon dioxide, as was also noted by DallaSanta et al. (2021) and Schirber et al. (2015).

Although greater Kelvin wave activity is an agreed-upon feature of a warming climate (Bartana et al., 2023; Raghavendra et al., 2019), changes in parameterized gravity waves are less certain. In comprehensive models, increased gravity wave activity generally occurs alongside greater tropical convective precipitation rates. For instance, the tropical-average precipitation rate and stratospheric gravity wave momentum flux in the QBOi CESM1-WACCM simulations increased by roughly 5% and 50%, respectively, due to a 4 K increase in SSTs (Richter et al., 2022). It is thus surprising that the CAM7 aqua-planet tropical convective precipitation rate did not respond appreciably to warmer SSTs, even while realistic changes in the intensity and depth of the parameterized deep convection occurred. In fact, when averaged between 10°N/S, the aqua-planet precipitation rate exhibited a slight decrease due to surface warming, in stark disagreement with the 2%–13% increase in the same metric predicted by the QBOi models (Richter et al., 2022). Additionally, changes in the stratospheric easterly gravity wave momentum flux were primarily due to changes in the background zonal wind. Although a similar mechanism was observed in a fully coupled CESM simulation (Lee et al., 2024), the magnitude of the zonal wind changes seen in the 4xCO2+4K aqua-planet case is much larger. It is possible that the unexpected behavior of the 4xCO2+4K tropical convection originates in the coupling between CAM7 physics and the SE dynamical core. A connection between dynamical core choice and the structure of tropical convection has already been established (Landu et al., 2014), and a recent aqua-planet study using version 6 of CAM (CAM6), which employs the finite volume (FV) dynamical core (Lin, 2004), found that a uniform 4 K increase in SSTs increased the deep tropical (5°S–5°N) precipitation rate by around 20% (Chien & Kim, 2024). Additionally, 4xCO2+4K aqua-planet simulations conducted using earlier versions of CAM with the FV dynamical core linked increasing tropical precipitation rates and greater gravity wave driving of the QBO. Thus, while the 4xCO2+4K aqua-planet QBO period decreases as expected in response to a greater flux of parameterized waves, inconsistent behavior in the deep convection and gravity wave schemes precludes the identification of realistic drivers of increased gravity wave momentum flux in a warmer climate.

Sensitivity studies of the parameterized gravity wave forcing of the QBO are also limited by uncertainties in the tuning parameters leveraged to align the simulated QBO with reality. For example, Schirber et al. (2015) present multiple atmosphere-only model simulations that differ only in the configuration of the tropical convective gravity wave scheme, yet project opposite QBO period shifts under a 4 K SST warming. Although the Beres et al. (2005) scheme is arguably one of the more physically grounded convective gravity wave parameterizations due to its interactive gravity wave source, even comparing favorably with high-resolution simulations of gravity waves in a warming climate (Franke et al., 2023), it still requires artificially scaled convective heating depths to produce a realistic QBO (Alexander et al., 2021). Given that the convective heating depth directly impacts the phase speeds of all gravity waves launched by the Beres et al. (2005) scheme, this artificial scaling could reduce the sensitivity of the aqua-planet QBO to increased SSTs and deepening tropical convection. Even if the shorter QBO period mediated by the Beres et al. (2005) scheme in fully coupled (Butchart et al., 2020; Lee et al., 2024), atmosphere-only (Richter et al., 2022; Schirber et al., 2015), and, in this study, aqua-planet simulations is the correct response to increased SSTs, the magnitude of that response cannot be accurately diagnosed while tuning uncertainties persist.

Only a handful of other aqua-planet models contain an internally generated QBO, and the aqua-planet framework has never been used to analyze the QBO response to climate change. That a realistic QBO which responded in a physically grounded manner to increased carbon dioxide concentrations and SSTs has been simulated in the CAM7 aqua-planet configuration suggests that aqua-planet models could be a useful framework for deciphering model disagreements with respect to QBO projections, similar to how aqua-planets have been used to compare models' handling of aerosols or cloud processes (Medeiros, 2020; Medeiros et al., 2008). A possible model intercomparison project could focus on differences in gravity wave parameterizations in an effort to minimize the uncertainty in the QBO response to climate change. In addition, the aqua-planet configuration could be used to provide insight into the mechanisms, or teleconnections, by which the QBO affects the global climate. Recent work has analyzed the climate of fixed-winter and summer aqua-planet simulations (Schemm & Röthlisberger, 2024), which exhibit seasonal patterns that would not otherwise appear in a typical aqua-planet. Since many of the QBO teleconnections are seasonally dependent (e.g., the Holton-Tan effect) and the aqua-planet configuration removes sources of variability that complicate the detection of teleconnection pathways (Anstey et al., 2022), a fixed-seasonal aqua-planet model may be useful for examining the mechanisms through which the QBO influences the tropospheric and stratospheric climate.

Appendix A: Model Parameters

This appendix describes the CAM7 model parameters that were changed to tune the QBO period. All of the parameters in Table A1 except for CF, which must be directly modified in the *gw_convect.F90* subroutine, are namelist parameters that can be specified in an input file to CAM. As described in Section 2.1, the convective gravity wave efficiency, *effgw_beres_dp*, and the convective fraction, CF, were changed from the default CAM7 settings to produce a more realistic QBO. The heating depth scaling (*gw_qbo_hdepth_scaling*) was not tuned, but is included in Table A1 for completeness. The other parameters in Table A1, which modify the SE dynamical core, the gravity wave drag scheme, cloud macrophysics and turbulence (CLUBB), dust emissions, and chlorofluorocarbon concentrations, were adjusted to match the CAM6 default settings. These choices were found to improve the Eliassen-Palm flux divergence body force in the stratosphere, which occasionally had the incorrect sign during westerly to easterly QBO transitions, but had little effect on the QBO amplitude and period.

Table A1
Differences in Model Parameters Between Default CAM7 and the Tuned Version of CAM7 Used in This Study

Scheme	Tuned parameter	Description	CAM7 default	New value
SE Dynamical Core	se_hypervis_subcycle_sponge	Sponge layer diffusion subcycles per dynamics step	3	1
	se_nu_top	Model top second-order viscosity (m^2 s^{-1})	1.0×10^6	1.25×10^5
Gravity Wave Drag	effgw_beres_dp	Convective gravity wave efficiency	0.70	0.60
	gw_qbo_hdepth_scaling	Scaling factor for convective heating depth	0.25	0.25
	use_gw_movmtn_pbl	Enable PBL moving mountains gravity waves	TRUE	FALSE
	gw_apply_tndmax	Limit maximum wind tendency from stress divergence	FALSE	TRUE
	CF	Percent of grid cell in which convection is occurring	20	10
Cloud Layers Unified By Binormals (CLUBB)	clubb_c7	Low skewness of buoyancy damping of water flux	0.1	0.5
	clubb_c8	Constant associated with the damping of $\overline{w'^3}$	4.5	4.2
	clubb_c_uu_shr	Variance of vertical velocity pressure terms coeff.	0.1	0.3
	clubb_detice_rad	Radius of detrained ice crystals (m)	61.0×10^{-6}	25.0×10^{-6}
	clubb_gamma_coef	Low skewness in gamma coefficient skewness function	0.3	0.308
	clubb_gamma_coefb	Limiting value of gamma when skewness of w is large	0.3	0.32
	clubb_l_intr_sfc_flux_smooth	Apply local u^* to momentum surface fluxes	TRUE	FALSE
	clubb_l_min_wp2_from_corr_wx	Set additional minimum threshold for variance of w	FALSE	TRUE
	clubb_l_predict_upwp_vpwp	Predict horizontal momentum fluxes	TRUE	FALSE
Dust Emission	dust_emis_fact	Tuning parameter for dust emissions	2.30	0.70
Greenhouse Gases	f11vmr	CFC11 volume mixing ratio (mol mol^{-1})	0.0	653.45×10^{-12}
	f12vmr	CFC12 volume mixing ratio (mol mol^{-1})	0.0	535.0×10^{-12}

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

CAM7 (tag: 6_4_070) is available for public use at https://github.com/ESCOMP/CAM (Danabasoglu et al., 2020). Zonal-mean, monthly mean CAM7 aqua-planet output data are available in a Zenodo repository, doi:10.5281/zenodo.21041889 (Johnson et al., 2026b). Version v0.1.0 of the Python and Jupyter Notebook scripts used to process the model output data and create figures is available under the MIT license on Github: https://github.com/aaroj11235/aqua-qbo, and is also archived on Zenodo, doi:10.5281/zenodo.18841132 (Johnson et al., 2026a).

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